# The Riemann–von Mangoldt clock and the zeros of the Riemann zeta function: an exact defect identity and variable prime cutoffs

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**Abstract**

Let $\gamma_k$ be the ordinates of the nontrivial zeros of the Riemann zeta function, counted with multiplicity, and let $\tau_k$ be the midpoint Riemann–von Mangoldt clock, defined by $N_0(\tau_k) = k - \frac{1}{2}$, where $N_0$ is the smooth part of the Riemann–von Mangoldt formula. *Known:* $\limsup_k |\tau_k - \gamma_k| \le 0.6331$ follows from the bound of Bellotti and Wong for the argument of the Riemann zeta function, and the ordinal displacement $s_k = k - \frac{1}{2} - N_0(\gamma_k)$ is asymptotically normal with variance $\frac{1}{2\pi^2}\log\log T$ by Selberg's formulae proved by Korolev. *New:* (A) for every $C^1$ model $F$ of the zero-counting function, sums of even powers of $k - \frac{1}{2} - F(\gamma_k)$ over consecutive ordinates equal weighted integrals of $N - F$ minus explicit sawtooth and boundary terms, with no hypothesis on the zeros; (B) if the clock equation is corrected by the Euler phase $S_X$ of the primes $p \le X$, the mean square over $T < \gamma_k \le 2T$ of the defect $u_k(X) = k - \frac{1}{2} - N_0(\gamma_k) - S_X(\gamma_k)$ is $\frac{1}{2\pi^2}\log\frac{\log T}{\log X} + O(1 + \sqrt{\log\frac{\log T}{\log X}})$ uniformly for $2 \le X \le T^{1/2}$, so the corrected clock places almost all zeros within a bounded number of positions if and only if $\log X \gg \log T$; (C) the mean-square gain of the computable one-step corrected clock on any finite set of zeros is, exactly, a diagonal prime-response term minus a cross term, which accounts for the cutoff that is optimal at a given height. Under two named hypotheses the prime response of $s_k$ at the zeros is $1 - \Phi(\theta)$, $\Phi(\theta) = (1+\theta)\log(1+\theta) + (1-\theta)\log(1-\theta)$, $\theta = \log p/\log(T/2\pi)$; a determinantal model motivating one hypothesis is kept separate. The observables of (B) and (C) are computed on the first 2 001 052 zeros of the Riemann zeta function.

*Keywords:* Riemann zeta function, zeros of the Riemann zeta function, Riemann–von Mangoldt formula, argument of the Riemann zeta function, truncated Euler product
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**Contents**

**1 Introduction** **2**
1.1 The Riemann–von Mangoldt clock and the ordinates . . . . . 2
1.2 What is known and only used . . . . . 3
1.3 What this paper adds . . . . . 3
1.4 Organization . . . . . 4

**2 Notation** **4**
2.1 List of symbols . . . . . 4
2.2 Counting functions . . . . . 6

2.3 The RvM clock and the displacements . . . . . 6
2.4 Euler phases and the corrected clock . . . . . 7

**3 External results** **8**

**4 Theorem A: an exact defect identity at the zeros** **9**

**5 Theorem B: how much prime information the clock needs** **11**
5.1 Lemmas on prime polynomials . . . . . 12
5.2 Proof of Theorem 5.1 . . . . . 13

**6 Theorem C: exact accounting for the one-step clock** **14**

**7 Computations** **16**
7.1 Data and methods . . . . . 16
7.2 The clock and the ordinates . . . . . 16
7.3 Direct defects (Theorem B) . . . . . 17
7.4 Exact accounting (Theorem C) . . . . . 17
7.5 The one-step clock against the cutoff . . . . . 17

**8 Conditional results** **20**
8.1 Fixed frequencies under Hypothesis L . . . . . 21
8.2 The response at the ordinates and its Fourier reduction . . . . . 22
8.3 The kernel $\mathfrak{K}_\theta$ . . . . . 24
8.4 The moving-prime law under Hypotheses U and W . . . . . 24

**9 Heuristic support and numerical evidence** **25**
9.1 A determinantal model whose connected kernel is $\mathfrak{K}_\theta$ . . . . . 25
9.2 Numerical evidence . . . . . 26

**10 Conclusions** **29**
10.1 What was known . . . . . 29
10.2 What this paper adds . . . . . 29
10.3 Why a finer knowledge of $\tau_k - \gamma_k$ matters . . . . . 30
10.4 Future work . . . . . 30

## 1. Introduction

*1.1. The Riemann–von Mangoldt clock and the ordinates*

Let $\zeta(s)$ be the Riemann zeta function, let $N(T)$ be the number of its nontrivial zeros $\rho = \beta + i\gamma$ with $0 < \gamma \le T$, counted with multiplicity, and list their ordinates as $\gamma_1 \le \gamma_2 \le \cdots$, with multiplicity. Away from ordinates the Riemann–von Mangoldt formula reads [22, §§9.3–9.4]

$$N(T) = N_0(T) + \frac{1}{\pi}\arg\zeta\Big(\frac{1}{2} + iT\Big) + O\Big(\frac{1}{T}\Big), \qquad N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}, \tag{1.1}$$

the argument of the Riemann zeta function being obtained by continuous variation along $2 \to 2+iT \to \frac{1}{2}+iT$ from $\arg\zeta(2)=0$. The *midpoint Riemann–von Mangoldt clock* (RvM clock) is the sequence $\tau_k$ defined by

$$N_0(\tau_k) = k - \frac{1}{2}, \qquad \tau_k = \frac{2\pi\,(k-\frac{11}{8})}{W_0\Big((k-\frac{11}{8})/e\Big)} \qquad (k \geq 1), \tag{1.2}$$

where $W_0$ is the principal branch of the Lambert function; the Lambert form appears in the work of França and LeClair [5, 6]. The clock is explicit and lies, in phase, half way between consecutive Gram points. This paper studies its relation to the ordinates $\gamma_k$ of the zeros of the Riemann zeta function: in which sense the clock describes them, and how much prime information has to be added to it. The relation is measured in units of zeros by the ordinal displacement and by its prime-corrected version, the *ordinal defect*,

$$s_k = k - \frac{1}{2} - N_0(\gamma_k), \qquad u_k(X) = s_k - S_X(\gamma_k), \qquad S_X(t) = -\frac{1}{\pi}\sum_{p\leq X} \arg\Big(1-p^{-1/2-it}\Big). \tag{1.3}$$

$u_k(X)$ measures how well the true ordinate satisfies the corrected clock equation $N_0(t) + S_X(t) = k - \frac{1}{2}$; $|u_k(X)| < \frac{1}{2}$ means that the corrected count assigns to $\gamma_k$ its correct index.

*1.2. What is known and only used*

The following facts are quoted with references in Section 3 and are not reproved.

- The displacement $\delta_k = \tau_k - \gamma_k$ is bounded, and $\limsup_k |\delta_k| \leq 2\pi \cdot 0.10076 = 0.6331\ldots$; this follows at once from the explicit bound of Bellotti and Wong for $N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e}$ [1].

- The ordinal displacement $s_k$ differs by less than 2 from minus the integer displacement of $\gamma_k$ with respect to the Gram intervals, whose even moments are Gaussian with variance $\frac{1}{2\pi^2}\log\log T$ and whose odd moments are of lower order, by Selberg's formulae as proved by Korolev [12, 11]; hence $s_k$ is asymptotically normal with this variance. Hence the bare clock misplaces almost all zeros by an unbounded number of positions, although $\delta_k \to 0$ in mean square.

- Selberg's approximation of $S(t) = \frac{1}{\pi}\arg\zeta(\frac{1}{2}+it)$ by prime sums up to a small power of $T$ holds in $L^4(T, 2T)$ [19, 20], in the form proved by Fujii [7, §3]. It concerns $t$-averages, not the ordinates.

- LeClair [14, Sect. III, VI] replaces the argument in the equation of the individual zero by a finite Euler product, asks for the number of primes to grow with the height, and computes with a fixed number of primes.

*1.3. What this paper adds*

Let $\mathcal{I}_T = \{k : T < \gamma_k \leq 2T\}$. Theorems A, B and C are unconditional. We have not found them in the literature; absence from a search is weak evidence and does not establish priority.

**A. Exact defect identity** (Theorem 4.1). For every real $C^1$ model $F$ of the counting function and $u_k = k - \frac{1}{2} - F(\gamma_k)$, the sum $\sum u_k^{2j}$ over consecutive ordinates equals $\int (N-F)^{2j}\, dF$ minus explicit lower-degree terms ($\frac{1}{12}$ per zero when $j=1$) and two boundary terms. Neither simplicity of the zeros, nor monotonicity of $F$, nor any hypothesis on the real parts $\beta_k$ is used. It converts $t$-averages into averages over the ordinates without short-interval zero counts and without error term.

**B. How much prime information the clock needs** (Theorem 5.1). Uniformly for $2 \le X \le T^{1/2}$, with $c = \log X / \log T$,

$$\frac{1}{\#\mathcal{I}_T} \sum_{k \in \mathcal{I}_T} u_k(X)^2 = \frac{1}{2\pi^2} \log \frac{1}{c} + O\Big(1 + \sqrt{\log(1/c)}\Big). \tag{1.4}$$

A fixed set of primes leaves a mean-square defect $\sim (2\pi^2)^{-1} \log\log T$; primes up to $T^c$ keep it bounded; boundedness holds if and only if $\liminf \log X / \log T > 0$ (Corollary 5.4). Thus the growth of the number of primes asked for in [14] is necessary but not sufficient.

**C. Exact accounting for the computable clock** (Theorem 6.1). For the one-step clock $\tau_k^X = \tau_k - S_X(\tau_k)/\mathcal{L}(\tau_k)$, the mean-square gain over any finite set of zeros equals a diagonal prime-response term minus a cross term, with no independence assumption, and the residual of $\tau_k^X$ differs from $u_k(X)$ by an explicit transport term. Computed on the data, the cross term grows with the cutoff at fixed height and decreases with the height at fixed cutoff, which explains the cutoff that is optimal at a given height (Section 7).

*Conditional results* (Section 8) are rigorous implications from named hypotheses: fixed-frequency limits of the prime responses under a Landau-type Hypothesis L, and the moving-prime law $1-\Phi(\theta)$ under Hypotheses U and W (Proposition 8.11), resting on an unconditional reduction to a pair correlation shifted by a prime frequency (Theorem 8.7). *Heuristic support* (Section 9) is kept apart: a determinantal model whose connected kernel integrates to $-2\Phi(\theta)/\theta$ motivates Hypothesis W, and the comparison of the conjectured law with the data is reported there, including its systematic deviations.

### *1.4. Organization*

Section 2 lists the notation and Section 3 the external results. Sections 4, 5 and 6 prove Theorems A, B and C. Section 7 computes their observables on the first 2 001 052 zeros of the Riemann zeta function. Sections 8 and 9 contain the conditional results and the heuristic support. Section 10 separates what is known from what is new and discusses the use of a finer knowledge of $\tau_k - \gamma_k$. All programs and outputs are in a public repository (Data availability).

## 2. Notation

### *2.1. List of symbols*

Every symbol has one meaning throughout. The letter $\mu$ is not used (it is reserved for the Möbius function), $\pi$ is only the number 3.14159…, and prime counts are written $\#\{p \le y\}$. Dummy variables of integration ($t$, $y$) are local.

| symbol | meaning |
|---|---|
| $\zeta(s)$ | the Riemann zeta function |
| $\rho_k = \beta_k + i\gamma_k$ | nontrivial zeros of the Riemann zeta function, $0 < \gamma_1 \le \gamma_2 \le \cdots$ with multiplicity |
| $\nu(\gamma)$ | number of zeros of the Riemann zeta function with ordinate $\gamma$ (multiplicities included) |
| $N(t)$, $N(t^{\pm})$ | number of zeros with $0 < \gamma \le t$; one-sided limits |
| $\vartheta$, $\Theta_0$, $\varpi$ | Riemann–Siegel theta function; $\Theta_0(t) = \frac{t}{2}\log\frac{t}{2\pi e} - \frac{\pi}{8}$; $\varpi = \vartheta - \Theta_0$ |
| $g_m$ | Gram points, $\vartheta(g_m) = m\pi$ |
| $N_0$, $\mathcal{L}$ | $N_0 = \Theta_0/\pi + 1$; $\mathcal{L} = N_0'$, $\mathcal{L}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$ |
| $S$, $S^*$, $G$ | $S(t) = \frac{1}{\pi}\arg\zeta(\frac{1}{2} + it)$; midpoint value at ordinates; $G = N - N_0 = S + \varpi/\pi$ |
| $\mathcal{I}_T$, $\langle\cdot\rangle_I$ | $\{k : T < \gamma_k \le 2T\}$; mean over $k$ in a finite index set $I$ |
| $\tau_k$, $\tau_x$, $W_0$ | RvM clock (2.5); the same for real $x \ge 1$; principal Lambert function |
| $s_k$, $\delta_k$, $\xi_k$ | $s_k = k - \frac{1}{2} - N_0(\gamma_k)$; $\delta_k = \tau_k - \gamma_k$; mean-value point in $s_k = \mathcal{L}(\xi_k)\delta_k$ |
| $\Delta_k$ | integer displacement of $\gamma_k$ with respect to the Gram intervals (Theorem 3.4) |
| $\mathfrak{d}_k$ | $s_k + \Delta_k$, $\lvert\mathfrak{d}_k\rvert < 2$ (Remark 3.10) |
| $\mathfrak{s}_T$ | $(\frac{1}{2\pi^2}\log\log T)^{1/2}$ (Remark 3.10) |
| $\Lambda$, $a_n$ | von Mangoldt function; $a_n = -\Lambda(n)/(\pi\sqrt{n}\log n)$ |
| $A$, $A_X$, $\mathcal{P}$ | set of prime powers with $\sum_{n\in A}\lvert a_n\rvert < \infty$; $A_X = \{p^m : p \le X\}$; finite set of primes |
| $S_A$, $S_X$, $S_{\mathcal{P}}$ | $\sum_{n\in A} a_n \sin(t\log n)$; $S_{A_X}$; $S_{\{p^m : p\in\mathcal{P}\}}$ |
| $v_A$, $v_X$, $v_{\mathcal{P}}$ | $\frac{1}{2}\sum_{n\in A} a_n^2$; $v_{A_X}$; $v_{\{p^m : p\in\mathcal{P}\}}$ |
| $\sigma_X$, $M_X$ | bounds for $\lvert S_X\rvert$ and $\lvert S_X'\rvert$, (2.10) |
| $F$, $F_X$ | a real $C^1$ model of $N$; $F_X = N_0 + S_X$ |
| $u_k$, $u_k(X)$ | $k - \frac{1}{2} - F(\gamma_k)$; $k - \frac{1}{2} - F_X(\gamma_k)$ |
| $\Psi_j$, $\kappa_j$ | sawtooth polynomial (4.2); $\kappa_j = \frac{(2j)!}{j!(2\pi)^{2j}}$ |
| $X$, $c$, $C$, $T_0$ | prime cutoff; $c = \log X/\log T$; absolute constants |
| $Y$, $Y_0$, $S_Y^{\mathrm{pr}}$ | polynomial length; $Y_0 = T^{1/160}$; $S_Y^{\mathrm{pr}}(t) = -\frac{1}{\pi}\sum_{p<Y} p^{-1/2}\sin(t\log p)$ |
| $\mathcal{J}$, $V$, $\mathcal{W}$, $\mathcal{W}_1$ | primes between $X$ and $Y_0$; $V = \sum_{p\in\mathcal{J}} p^{-1}$; $\mathcal{W} = S_{Y_0}^{\mathrm{pr}} - S_X$; its part over $\mathcal{J}$ (5.4) |
| $E_0$, $E_X$, $\mathcal{E}_2$ | $S - S_{Y_0}^{\mathrm{pr}}$; $S - S_X$; $\int_T^{2T} E_X^2\,dt$ |
| $z_n$, $\mathcal{D}$, $\mathcal{D}_1$, $\mathcal{D}_2$, $\eta_n$, $\eta_n'$ | coefficients and Dirichlet polynomials in Lemmas 5.8–5.9 |
| $\tau_k^X$, $\delta_k^X$, $r_k(X)$, $\chi_k$ | one-step clock; $\tau_k^X - \gamma_k$; $\mathcal{L}(\tau_k)\delta_k^X$; second-order error (6.3) |
| $\mathcal{R}_n^{\gamma}(I)$, $\mathcal{R}_n^{\tau}(I)$ | prime-power responses at the zeros and at the clock points (6.1) |
| $\mathrm{Cr}_A(I)$, $D_X(I)$ | cross term (6.2); diagonal term (6.5) |
| $\lambda$ | real multiplier of an Euler phase |
| $\varsigma_n$, $\mathbf{Q}$, $\mathbf{b}$, $\boldsymbol{\lambda}$, $\Upsilon_{\boldsymbol{\lambda}}$ | $\varsigma_n(k) = \sin(\tau_k\log n)$; Gram matrix; projection vector; coefficients; trigonometric correction (Remark 6.3) |
| $\mathrm{BW}(t)$ | $0.10076\log t + 0.24460\log\log t + 8.08344$ (Bellotti–Wong) |
| $t_k^X$ | root of $F_X(t) = k - \frac{1}{2}$ (Remark 5.6) |
| $\ell_n$ | $\log n$ |
| $\psi$, $t_1$, $t_2$ | test function and endpoints in Lemma 4.4 |

| symbol | meaning |
|---|---|
| $L_T$, $\mathcal{N}_T$, $\theta$ | $\log(T/2\pi)$; $TL_T/2\pi$; $\log p/L_T$ |
| $\Phi(\theta)$ | $(1+\theta)\log(1+\theta)+(1-\theta)\log(1-\theta)$ |
| $\mathcal{A}_{T,p}$, $\mathcal{S}_{T,p}$ | one-particle term and shifted correlation (8.3) |
| $\mathcal{H}_n(t)$ | $\int_{14}^t S(y)n^{iy}\,dy$ (proof of Theorem 8.4) |
| $f_T$, $\mathfrak{m}_T$, $Z_T$, $\mathcal{B}_T$ | $S\mathbf{1}_{[T,2T]}$; its distributional derivative; Fourier transform of $\mathfrak{m}_T$; scaled bilinear integral (8.9) |
| $\omega$, $\omega_0$, $\alpha$ | frequency variables; $\alpha=\omega/L_T$ |
| $\mathfrak{K}_\theta$, $h$ | kernel (8.10); $h(q)=\max(1-\lvert q\rvert,0)$ |
| $R_0$, $\mathfrak{n}$, $\Pi_{\mathfrak{n}}$, $q$, $\Xi_q$ | model: circle length, number of modes, projection, lattice frequency, linear statistic |
| $\iota_1$, $\iota_2$, $\mathsf{m}_i$ | tilt parameters; multiplication operators (Section 9.1) |
| $y_0$, $w_T$ | fixed threshold; threshold tending to infinity |
| $K$, $B$ | 2 001 052, the number of ordinates used; saturation level (Section 7.5) |

*2.2. Counting functions*

Let $\vartheta(t)=\Im\log\Gamma(\frac14+\frac{it}{2})-\frac{t}{2}\log\pi$ for $t>0$. For $t>0$ not an ordinate [22, Thms. 9.3–9.4],

$$N(t)=\frac{1}{\pi}\vartheta(t)+1+S(t),\qquad S(t)=\frac{1}{\pi}\arg\zeta\Big(\tfrac12+it\Big)=O(\log t),\tag{2.1}$$

with $\arg\zeta$ defined by continuous variation from 2 to $2+it$ and then horizontally to $\frac12+it$. At an ordinate we keep the right-continuous $N$ and define $S$ by (2.1); $S^*(\gamma)=\frac12(S(\gamma^-)+S(\gamma^+))$. Put

$$\Theta_0(t)=\frac{t}{2}\log\frac{t}{2\pi e}-\frac{\pi}{8},\qquad N_0(t)=\frac{1}{\pi}\Theta_0(t)+1,\qquad \varpi(t)=\vartheta(t)-\Theta_0(t).\tag{2.2}$$

By Stirling's formula $\varpi(t)=\frac{1}{48t}+O(t^{-3})$ and $\varpi'(t)=O(t^{-2})$; in particular $\lvert\varpi(t)\rvert/\pi<\frac{1}{100}$ for $t\ge14$. For all $t>0$,

$$N(t)=N_0(t)+S(t)+\frac{\varpi(t)}{\pi},\qquad \mathcal{L}(t)=N_0'(t)=\frac{1}{2\pi}\log\frac{t}{2\pi}.\tag{2.3}$$

Since $N(2T)-N(T)=N_0(2T)-N_0(T)+O(\log T)$,

$$\#\mathcal{I}_T=T\mathcal{L}(T)+O(T).\tag{2.4}$$

*2.3. The RvM clock and the displacements*

$N_0$ is $C^\infty$ and strictly increasing on $(2\pi,\infty)$, with $N_0(2\pi)=-\frac18$ and $N_0(t)\to\infty$. Hence for every real $x\ge1$ the equation $N_0(t)=x-\frac12$ has exactly one solution $\tau_x>2\pi$. Writing $t=2\pi y$, the equation is $y\log(y/e)=x-\frac{11}{8}\ge-\frac38$; its root $y>1$ is $y=(x-\frac{11}{8})/W_0((x-\frac{11}{8})/e)$, with $y=e$ if $x=\frac{11}{8}$ [3]. Thus

$$\tau_x=\frac{2\pi\,(x-\frac{11}{8})}{W_0\Big((x-\frac{11}{8})/e\Big)},\qquad \frac{d\tau_x}{dx}=\frac{1}{\mathcal{L}(\tau_x)}=\frac{2\pi}{\log(\tau_x/2\pi)},\tag{2.5}$$

the derivative by implicit differentiation. Since $\Theta_0(\tau_k)=\pi(k-\frac32)$ and $\varpi=O(1/t)$, $\tau_k$ lies in phase half way between $g_{k-2}$ and $g_{k-1}$, up to $O(\tau_k^{-1}(\log\tau_k)^{-1})$.

**Definition 2.1.** For $k \geq 1$, $s_k = k - \frac{1}{2} - N_0(\gamma_k)$ and $\delta_k = \tau_k - \gamma_k$.

Since $\gamma_1 > 14 > 2\pi$ and $\tau_k > 2\pi$, the mean value theorem applied to $s_k = N_0(\tau_k) - N_0(\gamma_k)$ gives

$$s_k = \mathcal{L}(\xi_k)\,\delta_k \qquad \text{for some } \xi_k \text{ between } \gamma_k \text{ and } \tau_k. \tag{2.6}$$

**Lemma 2.2.** *For every $k \geq 1$,*

$$N(\gamma_k^-) - N_0(\gamma_k) + \frac{1}{2} \leq s_k \leq N(\gamma_k^+) - N_0(\gamma_k) - \frac{1}{2}, \tag{2.7}$$

*and $|s_k| \leq \max(|S(\gamma_k^-)|, |S(\gamma_k^+)|) + \frac{1}{2} + \frac{1}{100}$.*

*Proof.* The indices $j$ with $\gamma_j = \gamma_k$ are exactly $N(\gamma_k^-) + 1, \dots, N(\gamma_k^+)$, so $N(\gamma_k^-) + 1 \leq k \leq N(\gamma_k^+)$; subtract $\frac{1}{2} + N_0(\gamma_k)$. The second statement follows from (2.3) at $\gamma_k^{\pm}$, continuity of $N_0$ and $\varpi$, and $|\varpi|/\pi < \frac{1}{100}$. □

*2.4. Euler phases and the corrected clock*

For $n \geq 2$ let $a_n = -\Lambda(n)/(\pi\sqrt{n}\log n)$. For a set $A$ of prime powers with $\sum_{n\in A} |a_n| < \infty$,

$$S_A(t) = \sum_{n\in A} a_n \sin(t\log n), \qquad v_A = \frac{1}{2}\sum_{n\in A} a_n^2; \tag{2.8}$$

the series converges absolutely and uniformly in $t$, so $S_A$ is continuous and bounded. For $X \geq 2$, $\sum_{n\in A_X} |a_n| \leq \frac{1}{\pi}\sum_{p\leq X}\sum_{m\geq 1}\frac{1}{mp^{m/2}} < \infty$ and, since $\arg(1-z) = -\Im\sum_{m\geq 1} z^m/m$ for $|z| < 1$ with the principal argument,

$$S_X(t) = -\frac{1}{\pi}\sum_{p\leq X} \arg\bigl(1 - p^{-1/2-it}\bigr) = -\frac{1}{\pi}\sum_{p\leq X}\sum_{m\geq 1}\frac{\sin(mt\log p)}{m\,p^{m/2}} = S_{A_X}(t). \tag{2.9}$$

The sum of principal arguments is a continuous argument of the finite Euler product, not necessarily its principal argument. The series of derivatives converges uniformly, because $\sum_{p\leq X}\log p\sum_{m\geq 1} p^{-m/2} < \infty$; hence $S_X \in C^\infty(\mathbb{R})$, its derivatives may be taken termwise, and

$$|S_X(t)| \leq \sigma_X := \frac{1}{\pi}\sum_{p\leq X}\sum_{m\geq 1}\frac{1}{m\,p^{m/2}}, \qquad |S_X'(t)| \leq M_X := \frac{1}{\pi}\sum_{p\leq X}\frac{\log p}{\sqrt{p}-1}, \tag{2.10}$$

with $\sigma_X \ll \sqrt{X}/\log X$ and $M_X \ll \sqrt{X}$ by the prime number theorem.

**Definition 2.3.** For $X \geq 2$ let $F_X = N_0 + S_X$ and $u_k(X) = k - \frac{1}{2} - F_X(\gamma_k) = s_k - S_X(\gamma_k)$. The one-step clock and its residual are

$$\tau_k^X = \tau_k - \frac{S_X(\tau_k)}{\mathcal{L}(\tau_k)}, \qquad \delta_k^X = \tau_k^X - \gamma_k, \qquad r_k(X) = \mathcal{L}(\tau_k)\,\delta_k^X. \tag{2.11}$$

Since $F_X(\tau_k) - (k - \frac{1}{2}) = S_X(\tau_k)$, $\tau_k^X$ is the first step of $t \mapsto t - (F_X(t) - k + \frac{1}{2})/\mathcal{L}(t)$ from $t = \tau_k$. At a simple ordinate, (2.3) gives $N_0(\gamma_k) + S^*(\gamma_k) + \varpi(\gamma_k)/\pi = k - \frac{1}{2}$, so $u_k(X) = S^*(\gamma_k) - S_X(\gamma_k) + \varpi(\gamma_k)/\pi$ is the error of the prime approximation of the argument of the Riemann zeta function at the zero.

## 3. External results

The results of this section are known; they are used, not reproved.

**Theorem 3.1** ([22, Thms. 9.4, 9.9 and §9.12], [15]). *$S(t) = O(\log t)$; $\gamma_{k+1} - \gamma_k$ is bounded; and $S_1(t) = \int_0^t S(y)\,dy = O(\log t)$.*

**Theorem 3.2** (Bellotti–Wong [1]). *For $T \geq e$, $\Big|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e}\Big| \leq \mathrm{BW}(T)$.*

Earlier explicit versions are due to Trudgian [23] and to Hasanalizade, Shen and Wong [10].

**Theorem 3.3** (Selberg [19]; see [7, p. 225]). *For every integer $j \geq 1$, $\int_T^{2T} S(t)^{2j}\,dt = \kappa_j\, T(\log\log T)^j + O_j\Big(T(\log\log T)^{j-1/2}\Big)$.*

**Theorem 3.4** (Selberg; Korolev [12], [11, Thm. 6]). *Let $\Delta_k$ be the integer such that $g_{k+\Delta_k-1} < \gamma_k \leq g_{k+\Delta_k}$, and let $0 < \varepsilon_0 < 10^{-3}$. For every fixed $j \geq 1$, as $n_0 \to \infty$, uniformly for $n_0^{27/82+0.9\varepsilon_0} \leq h_0 \leq n_0^{27/82+\varepsilon_0}$,*

$$\sum_{n_0<k\leq n_0+h_0} \Delta_k^{2j} = \kappa_j h_0(\log\log n_0)^j\Big(1+o(1)\Big), \qquad \Big|\sum_{n_0<k\leq n_0+h_0} \Delta_k^{2j-1}\Big| \ll_j h_0(\log\log n_0)^{j-1}. \tag{3.1}$$

**Theorem 3.5** (Selberg's approximation in the form of Fujii [7, §3, pp. 232–233]). *$\int_T^{2T}\Big(S(t) - S_{Y_0}^{\mathrm{pr}}(t)\Big)^4 dt \ll T$, where $Y_0 = T^{1/160}$.*

In [7, §3] Selberg's difference between $\pi^{-1}\arg L(\frac12+it,\chi)$ and a prime sum is written as five explicit terms, each bounded on p. 233 in $L^{2k}(T,T+H)$ by $\ll k^2H^{1/2k}$ under the hypotheses of Fujii's Main Theorem. For the principal character (so that the $L$-function is the Riemann zeta function), $k=2$, $H=T$ and Fujii's parameter $a_2 = \frac14$, the prime sum runs over $p < T^{3/480} = Y_0$, and $\frac{1}{\pi}\Im p^{-1/2-it} = -\frac{1}{\pi}p^{-1/2}\sin(t\log p)$.

**Theorem 3.6** (Montgomery–Vaughan [17, Cor. 3]). *For complex $z_n$, $\int_T^{2T}\Big|\sum_{n\leq Y} z_n n^{-it}\Big|^2 dt = \sum_{n\leq Y}|z_n|^2\Big(T + O(n)\Big)$ with an absolute implied constant.*

(Replace $z_n$ by $z_n n^{-iT}$ to pass from $[0,T]$ to $[T,2T]$.)

**Theorem 3.7** (Mertens [16]). *For $y \geq 2$, $\sum_{p\leq y}\frac{\log p}{p} = \log y + O(1)$ and $\sum_{p\leq y}\frac1p = \log\log y + O(1)$.*

**Theorem 3.8** (Kusmin–Landau [9, Thm. 2.1]). *Let $\Omega$ be real with continuous monotone derivative on $[y_1,y_2]$ and $\|\Omega'\| \geq 1/H_{\mathrm{KL}} > 0$ there, where $\|\cdot\|$ is the distance to the nearest integer. Then $\sum_{y_1<n\leq y_2} e^{2\pi i\Omega(n)} \ll H_{\mathrm{KL}}$.*

**Theorem 3.9** (Landau [13]; uniform version Gonek [8]). *For fixed integer $n \geq 2$, $\sum_{0<\gamma\leq t} n^{\rho} = -\frac{t}{2\pi}\Lambda(n) + O(\log t)$, zeros counted with multiplicity.*

*Remark* 3.10 (Known consequences for the clock). (a) *Moments and law of $s_k$.* Since $\vartheta(\gamma_k)/\pi \in (k+\Delta_k-1, k+\Delta_k]$ and $\vartheta/\pi = N_0 - 1 + \varpi/\pi$, one has $s_k + \Delta_k = k + \Delta_k - \frac{1}{2} - N_0(\gamma_k) \in [-\frac{3}{2} + \frac{\varpi(\gamma_k)}{\pi}, -\frac{1}{2} + \frac{\varpi(\gamma_k)}{\pi})$, so $s_k = -\Delta_k + \mathfrak{d}_k$ with $|\mathfrak{d}_k| < 2$. The index set $\mathcal{I}_T$ is the integer interval $(N(T), N(2T)]$, of length $\asymp T \log T$. Cover it by consecutive windows $(n_0, n_0 + h_0]$ with $h_0 = \lfloor n_0^{27/82 + 0.95\varepsilon_0} \rfloor$, the last one shifted to end at $N(2T)$, so that it overlaps the previous one in at most $h_0 = O(T^{-1/2} \# \mathcal{I}_T)$ indices; in each window $\log\log n_0 = \log\log T + o(1)$. Summing (3.1) over the windows, and bounding the doubly counted indices by one window sum (for odd powers, by Hölder's inequality and the even moments), gives $\langle \Delta_k^{2j} \rangle_{\mathcal{I}_T} = \kappa_j (\log\log T)^j (1 + o(1))$ and $\langle \Delta_k^{2j-1} \rangle_{\mathcal{I}_T} = O_j((\log\log T)^{j-1})$. Expanding $s_k^r = (-\Delta_k + \mathfrak{d}_k)^r$ and bounding $\langle |\Delta_k|^{r-i} \rangle_{\mathcal{I}_T} \le \langle \Delta_k^{2\lceil r/2 \rceil} \rangle_{\mathcal{I}_T}^{(r-i)/2\lceil r/2 \rceil} \ll (\log\log T)^{(r-i)/2}$ for $1 \le i \le r$, one obtains for every fixed $j \ge 1$

$$\langle s_k^{2j} \rangle_{\mathcal{I}_T} = \kappa_j (\log\log T)^j \Big(1 + o(1)\Big), \qquad \langle s_k^{2j-1} \rangle_{\mathcal{I}_T} = O_j\Big((\log\log T)^{j-1}\Big). \tag{3.2}$$

With $\mathfrak{s}_T^2 = \frac{\log\log T}{2\pi^2}$, so that $\kappa_j (\log\log T)^j = \frac{(2j)!}{j! 2^j} \mathfrak{s}_T^{2j}$, the normalized moments $\langle (s_k/\mathfrak{s}_T)^r \rangle_{\mathcal{I}_T}$ therefore tend to the moments of a standard normal variable $Z$ for every $r \ge 1$: the even ones by the first relation, the odd ones to 0 because $(\log\log T)^{j-1}/\mathfrak{s}_T^{2j-1} \ll_j (\log\log T)^{-1/2}$. The even moments alone determine only the law of $|s_k|$; it is the odd-moment bound in (3.1) that gives the sign. The normal law is determined by its moments, so the method of moments [2, Thm. 30.2] shows that the empirical law of $s_k/\mathfrak{s}_T$ on $\mathcal{I}_T$ tends to that of $Z$. Only consequences for $|s_k|$ are used in this paper: $\langle s_k^2 \rangle_{\mathcal{I}_T} \sim \frac{1}{2\pi^2} \log\log T$, $\limsup_k |s_k| = \infty$, and $\#\{k \in \mathcal{I}_T : |s_k| \le y_0\} = o(\#\mathcal{I}_T)$ for every fixed $y_0 > 0$; for these the even moments suffice.

(b) *Bound for $\delta_k$.* Theorem 3.2, applied as $T \to \gamma_k^{\pm}$ (both sides have one-sided limits), gives $|N(\gamma_k^{\pm}) - N_0(\gamma_k) + \frac{7}{8}| \le \mathrm{BW}(\gamma_k)$, and (2.7) gives $-\mathrm{BW}(\gamma_k) - \frac{3}{8} \le s_k \le \mathrm{BW}(\gamma_k) - \frac{11}{8}$. As $\mathcal{L}$ is positive and increasing on $(2\pi, \infty)$ and $s_k = \int_{\gamma_k}^{\tau_k} \mathcal{L}$,

$$-\frac{\mathrm{BW}(\gamma_k) + 3/8}{\mathcal{L}(\tau_k)} \le \delta_k \le \frac{\mathrm{BW}(\gamma_k) - 11/8}{\mathcal{L}(\gamma_k)} \qquad (k \ge 1). \tag{3.3}$$

Hence $\delta_k = O(1)$, $\mathcal{L}(\tau_k)/\mathcal{L}(\gamma_k) \to 1$, and $\limsup_k |\delta_k| \le 2\pi \cdot 0.10076 = 0.6331\ldots$, an immediate consequence of [1]. A limit superior is not a supremum: $|\delta_{34}| = 0.993711$.

## 4. Theorem A: an exact defect identity at the zeros

**Theorem 4.1.** *Let $1 \le k_1 < k_2$ be indices, $F$ a real $C^1$ function on an open interval containing $[\gamma_{k_1}, \gamma_{k_2}]$, and $u_k = k - \frac{1}{2} - F(\gamma_k)$ for $k_1 \le k \le k_2$. For every integer $j \ge 1$,*

$$\int_{\gamma_{k_1}}^{\gamma_{k_2}} \Big(N(t) - F(t)\Big)^{2j} F'(t)\, dt = \sum_{k=k_1+1}^{k_2-1} \Psi_j(u_k) + \frac{(\frac{1}{2} + u_{k_1})^{2j+1} + (\frac{1}{2} - u_{k_2})^{2j+1}}{2j+1}, \tag{4.1}$$

*where*

$$\Psi_j(u) = \frac{(\frac{1}{2} + u)^{2j+1} + (\frac{1}{2} - u)^{2j+1}}{2j+1} = \sum_{l=0}^{j} \binom{2j+1}{2l} \frac{4^{l-j}}{2j+1}\, u^{2l} = u^{2j} + \sum_{l<j} \binom{2j+1}{2l} \frac{4^{l-j}}{2j+1} u^{2l}. \tag{4.2}$$

*In particular, since* $\Psi_1(u) = u^2 + \frac{1}{12}$,

$$\sum_{k=k_1+1}^{k_2-1} u_k^2 = \int_{\gamma_{k_1}}^{\gamma_{k_2}} \Big(N(t) - F(t)\Big)^2 F'(t)\,dt - \frac{k_2 - k_1 - 1}{12} - \frac{(\frac{1}{2} + u_{k_1})^3 + (\frac{1}{2} - u_{k_2})^3}{3}. \tag{4.3}$$

*Proof. Existence.* On $[\gamma_{k_1}, \gamma_{k_2}]$ the function $N$ is a step function with finitely many jumps (Theorem 3.1) and $F$, $F'$ are continuous; the integrand is bounded and piecewise continuous, hence Riemann integrable.

*One gap.* Fix $k$ with $k_1 \le k < k_2$. If $\gamma_k < \gamma_{k+1}$, then $N(t) = k$ for $t \in (\gamma_k, \gamma_{k+1})$, because exactly $k$ ordinates counted with multiplicity are $\le t$. The function $t \mapsto -(k - F(t))^{2j+1}/(2j+1)$ is $C^1$ with derivative $(k - F(t))^{2j} F'(t)$ (chain rule; no monotonicity of $F$ is needed), so

$$\begin{aligned} \int_{\gamma_k}^{\gamma_{k+1}} \Big(k - F(t)\Big)^{2j} F'(t)\,dt &= \frac{(k - F(\gamma_k))^{2j+1} - (k - F(\gamma_{k+1}))^{2j+1}}{2j+1} \\ &= \frac{(\frac{1}{2} + u_k)^{2j+1} + (\frac{1}{2} - u_{k+1})^{2j+1}}{2j+1}, \end{aligned} \tag{4.4}$$

because $k - F(\gamma_k) = \frac{1}{2} + u_k$ and $k - F(\gamma_{k+1}) = -\frac{1}{2} + u_{k+1}$. If $\gamma_k = \gamma_{k+1}$, the left side of (4.4) is 0; then $u_{k+1} = u_k + 1$, and the right side equals $((\frac{1}{2} + u_k)^{2j+1} + (-\frac{1}{2} - u_k)^{2j+1})/(2j+1) = 0$. So (4.4) holds for every $k$, and the values of $N$ at the finitely many ordinates do not affect the integrals.

*Summation and boundary terms.* Summing (4.4) over $k_1 \le k < k_2$ (a finite sum), each $k$ with $k_1 < k < k_2$ contributes $(\frac{1}{2} + u_k)^{2j+1}$ from the gap starting at $\gamma_k$ and $(\frac{1}{2} - u_k)^{2j+1}$ from the gap ending at $\gamma_k$, that is $\Psi_j(u_k)$; the unmatched terms are $(\frac{1}{2} + u_{k_1})^{2j+1}$ and $(\frac{1}{2} - u_{k_2})^{2j+1}$. In the binomial expansion of $(\frac{1}{2} + u)^{2j+1} + (\frac{1}{2} - u)^{2j+1}$ the odd powers of $u$ cancel and $u^{2l}$ has coefficient $2\binom{2j+1}{2l} 2^{2l-2j-1}$, which gives (4.2); the leading coefficient is $\binom{2j+1}{2j}/(2j+1) = 1$. □

*Remark* 4.2. $\frac{1}{12}$ is the mean square of a centred unit sawtooth: between consecutive zeros $N - F$ decreases through one unit in the variable $F(t)$. If $F'$ changes sign, (4.3) remains exact with the integral counted with sign. With $F = N_0$ and Theorem 3.3, induction on $j$ through the lower-degree terms of $\Psi_j$ recovers the even moments in (3.2) without Gram intervals; as they are known, we do not write this out. Theorem 4.1 involves only even powers and gives no information on the odd moments, hence none on the sign of $s_k$. The purpose of Theorem 4.1 is Theorem 5.1, where $F = F_X$ need not be monotone and no Gram-interval structure is available.

**Corollary 4.3.** *For* $X \ge 2$ *and* $1 \le k_1 < k_2$,

$$\begin{aligned} \sum_{k=k_1+1}^{k_2-1} u_k(X)^2 = \int_{\gamma_{k_1}}^{\gamma_{k_2}} \Big(S - S_X + \frac{\varpi}{\pi}\Big)^2 \Big(\mathcal{L} + S_X'\Big) dt \\ - \frac{k_2 - k_1 - 1}{12} - \frac{(\frac{1}{2} + u_{k_1}(X))^3 + (\frac{1}{2} - u_{k_2}(X))^3}{3}. \end{aligned} \tag{4.5}$$

*Proof.* $F_X$ is $C^\infty$ on $(0, \infty)$ (Section 2.4), $F_X' = \mathcal{L} + S_X'$, and $N - F_X = S - S_X + \varpi/\pi$ by (2.3); apply Theorem 4.1 with $j = 1$. □

**Lemma 4.4** (First-moment identity)**.** *Let $2\pi < t_1 < t_2$ be points which are not ordinates and $\psi \in C^1[t_1, t_2]$, real or complex. Then*

$$\sum_{t_1<\gamma_k\le t_2} s_k\psi(\gamma_k) = \int_{t_1}^{t_2} G(t)\psi(t)\mathcal{L}(t)\,dt + \frac{1}{2}\Big[\psi\, G^2\Big]_{t_1}^{t_2} - \frac{1}{2}\int_{t_1}^{t_2} G(t)^2\psi'(t)\,dt. \tag{4.6}$$

*Proof.* On $[t_1, t_2]$ there are finitely many ordinates, none at $t_1$ or $t_2$. The function $G = N - N_0$ is right-continuous and of bounded variation, $C^\infty$ with $G' = -\mathcal{L}$ between ordinates, and it jumps by $\nu(\gamma)$ at an ordinate $\gamma$. Let $G^*(\gamma) = \frac{1}{2}(G(\gamma^-) + G(\gamma^+))$. The indices with $\gamma_k = \gamma$ are $N(\gamma^-) + 1, \dots, N(\gamma^-) + \nu(\gamma)$, so $\sum_{\gamma_k=\gamma}(k - \frac{1}{2}) = \nu(\gamma)N(\gamma^-) + \frac{\nu(\gamma)^2}{2} = \nu(\gamma)\frac{N(\gamma^-)+N(\gamma^+)}{2}$ and, $N_0$ being continuous, $\sum_{\gamma_k=\gamma} s_k = \nu(\gamma)G^*(\gamma)$. The left side of (4.6) therefore equals the finite sum $\sum_{t_1<\gamma<t_2} \nu(\gamma)G^*(\gamma)\psi(\gamma)$ over distinct ordinates. As Lebesgue–Stieltjes measures on $(t_1, t_2]$,

$$d(G^2) = -2G\mathcal{L}\,dt + \sum_{t_1<\gamma<t_2} 2G^*(\gamma)\nu(\gamma)\,\delta_\gamma, \tag{4.7}$$

because $(G^2)' = 2GG' = -2G\mathcal{L}$ between ordinates and $G(\gamma^+)^2 - G(\gamma^-)^2 = (G(\gamma^+) + G(\gamma^-))\nu(\gamma) = 2G^*(\gamma)\nu(\gamma)$. Integrating the bounded Borel function $\psi$ against (4.7) gives $\sum_{t_1<\gamma<t_2} \nu(\gamma)G^*(\gamma)\psi(\gamma) = \frac{1}{2}\int_{(t_1,t_2]} \psi\, d(G^2) + \int_{t_1}^{t_2} G\psi\mathcal{L}\,dt$. Since $\psi$ is $C^1$ and $G^2$ is right-continuous of bounded variation and continuous at $t_1$ and $t_2$, integration by parts for Lebesgue–Stieltjes integrals [4, §3.5] gives $\int_{(t_1,t_2]} \psi\, d(G^2) = [\psi G^2]_{t_1}^{t_2} - \int_{t_1}^{t_2} G^2\psi'\,dt$. All integrands are bounded and measurable, so all integrals exist. □

## 5. Theorem B: how much prime information the clock needs

**Theorem 5.1.** *There are absolute constants $T_0$ and $C$ such that for $T \ge T_0$ and every $X$ with $2 \le X \le T^{1/2}$, writing $c = \log X/\log T$,*

$$\Big|\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 - \frac{1}{2\pi^2}\log\frac{1}{c}\Big| \le C\Big(1 + \sqrt{\log(1/c)}\Big). \tag{5.1}$$

Here $\log(1/c) = \log\log T - \log\log X \in [\log 2, \log\log T - \log\log 2]$. For fixed $c$ both the main term and the error are bounded; the theorem does not identify a limit at fixed $c$. It concerns the defect at the true ordinate and does not select a root of $F_X(t) = k - \frac{1}{2}$.

**Corollary 5.2** (fixed cutoff)**.** *For fixed $X \ge 2$, $\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \frac{1}{2\pi^2}\log\log T + O_X(\sqrt{\log\log T})$, and $\#\{k \in \mathcal{I}_T : |u_k(X)| \le y_0\} = o(\#\mathcal{I}_T)$ for every fixed $y_0 > 0$.*

*Proof.* The first statement is (5.1). Since $|u_k(X) - s_k| \le \sigma_X$, $|u_k(X)| \le y_0$ implies $|s_k| \le y_0 + \sigma_X$; apply Remark 3.10(a). □

**Corollary 5.3** (power cutoff)**.** *For fixed $0 < c \le \frac{1}{2}$ and $X = T^c$,*

$$\limsup_{T\to\infty} \frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(T^c)^2 \le \frac{1}{2\pi^2}\log\frac{1}{c} + C\Big(1 + \sqrt{\log(1/c)}\Big), \tag{5.2}$$

*and $\#\{k \in \mathcal{I}_T : |u_k(T^c)| > w_T\} = o(\#\mathcal{I}_T)$ whenever $w_T \to \infty$. The same holds for $X = N(T)^c$ with fixed $0 < c < \frac{1}{2}$, since then $X = T^{c'}$ with $c' = c + O(\log\log T/\log T)$.*

*Proof.* (5.1) and Chebyshev's inequality. □

**Corollary 5.4** (dichotomy)**.** *Let $X = X(T)$ with $2 \le X \le T^{1/2}$. The mean square of $u_k(X)$ over $\mathcal{I}_T$ is bounded as $T \to \infty$ if and only if $\liminf_{T\to\infty} \log X/\log T > 0$. For fixed $X_0 \ge 2$ and $c \in (0, \frac12]$,*

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(u_k(X_0)^2 - u_k(T^c)^2\Big) = \frac{1}{2\pi^2}\log\log T + O\Big(\sqrt{\log\log T}\Big). \tag{5.3}$$

*Proof.* Both directions follow from (5.1), because $y \mapsto \frac{y}{2\pi^2} \pm C(1+\sqrt{y})$ is bounded on bounded sets and tends to infinity; (5.3) is the difference of the two cases of (5.1). □

*Remark* 5.5. Since $\log \#\{p \le X\} = \log X + O(\log\log X)$, the condition of Corollary 5.4 says that the logarithm of the number of primes must grow proportionally to $\log T$. A fixed number of primes, as in the computations of [14, Sect. VI], leaves almost all zeros violating the corrected equation by more than any fixed amount; the requirement that the number of primes tend to infinity [14, Sect. III] is necessary but not sufficient. The range $c > \frac12$ is not covered. The cutoff is constant on each window $\mathcal{I}_T$; with $T = 2^r$ and $X = 2^{rc}$ on $\mathcal{I}_{2^r}$ it depends on the ordinal through its window, but a cutoff varying at every ordinate is not covered.

*Remark* 5.6 (Localization of the ordinates). If $M_X \le \frac12\mathcal{L}(T)$, then $F_X' \ge \mathcal{L} - M_X \ge \frac12\mathcal{L}(T)$ on $[T, 2T]$, so $F_X$ is strictly increasing there. For $k \in \mathcal{I}_T$ with $F_X(T) \le k - \frac12 \le F_X(2T)$ the equation $F_X(t) = k - \frac12$ has exactly one root $t_k^X \in [T, 2T]$ (intermediate value theorem), and $u_k(X) = F_X(t_k^X) - F_X(\gamma_k)$ gives $|\gamma_k - t_k^X| \le 2|u_k(X)|/\mathcal{L}(T)$ (mean value theorem). As $M_X \asymp \sqrt{X}$, this applies only for $X \ll \log^2 T$. For $X = T^c$, $F_X$ need not be monotone, and Theorem 5.1 controls the equation at the true ordinates without selecting a root.

*5.1. Lemmas on prime polynomials*

Fix $T \ge T_0$ and $2 \le X \le T^{1/2}$, let $Y_0 = T^{1/160}$, and split $S_X = S_X^{(1)} + S_X^{(2)} + S_X^{(\ge3)}$ according to $m = 1$, $m = 2$, $m \ge 3$ in (2.9); the three series converge absolutely and uniformly. Let $\mathcal{J}$ be the set of primes in $(X, Y_0)$ if $X < Y_0$ and in $[Y_0, X]$ if $X \ge Y_0$, and put

$$\begin{aligned} \mathcal{W}_1(t) &= -\frac{1}{\pi}\sum_{p\in\mathcal{J}}\frac{\sin(t\log p)}{\sqrt{p}}, \qquad V = \sum_{p\in\mathcal{J}}\frac{1}{p}, \\ \mathcal{W} &= S_{Y_0}^{\mathrm{pr}} - S_X = \pm\mathcal{W}_1 - S_X^{(2)} - S_X^{(\ge3)}, \end{aligned} \tag{5.4}$$

with $+$ if $X < Y_0$ and $-$ if $X \ge Y_0$. The last identity holds because $S_X^{(1)}$ is the sum of $-\frac{1}{\pi}p^{-1/2}\sin(t\log p)$ over the primes $p \le X$, while $S_{Y_0}^{\mathrm{pr}}$ is the same sum over $p < Y_0$.

**Lemma 5.7.** *$V = \log(1/c) + O(1)$ uniformly for $\log 2/\log T \le c \le \frac12$.*

*Proof.* By Theorem 3.7, $V = |\log\log Y_0 - \log\log X| + O(1) = |\log\frac{1}{160c}| + O(1)$. For $c \le \frac{1}{160}$ this is $\log\frac1c - \log 160 + O(1)$; for $\frac{1}{160} < c \le \frac12$ both $V$ and $\log\frac1c$ are $O(1)$. □

**Lemma 5.8.** *For real $z_n$, $2 \le n \le Y$,*

$$\int_T^{2T}\Big(\sum_{n\le Y} z_n\sin(t\log n)\Big)^2 dt = \frac{T}{2}\sum_{n\le Y} z_n^2 + O\Big(\sum_{n\le Y} n z_n^2 + \Big(\sum_{n\le Y}|z_n|\Big)^2\Big), \tag{5.5}$$

*and the same with* cos *in place of* sin.

*Proof.* With the finite sum $\mathcal{D}(t)=\sum z_n n^{-it}$, $\sum z_n\sin(t\log n)=-\Im\mathcal{D}$ and $(\Im\mathcal{D})^2=\frac12(|\mathcal{D}|^2-\Re\mathcal{D}^2)$. Theorem 3.6 evaluates $\int|\mathcal{D}|^2$, and $|\int_T^{2T}(n_1n_2)^{-it}dt|\le 2/\log(n_1n_2)\le 2/\log 4$ gives $|\int\mathcal{D}^2|\le(2/\log 4)(\sum|z_n|)^2$. For cos use $\Re\mathcal{D}$ and $(\Re\mathcal{D})^2=\frac12(|\mathcal{D}|^2+\Re\mathcal{D}^2)$. □

**Lemma 5.9.** *With absolute implied constants,*

$$\int_T^{2T}\mathcal{W}^2\,dt=\frac{T}{2\pi^2}V+O\Big(T(1+\sqrt{V})\Big),\tag{5.6}$$

$$\int_T^{2T}\mathcal{W}^4\,dt\ll T(1+V)^2,\tag{5.7}$$

$$\int_T^{2T}S_X'(t)^2\,dt\ll T(1+\log X)^2.\tag{5.8}$$

*Proof.* All primes involved are $\le\max(X,Y_0)\le T^{1/2}$. Lemma 5.8 with $z_p=\frac{1}{\pi\sqrt{p}}$ ($p\in\mathcal{J}$) gives $\int\mathcal{W}_1^2=\frac{T}{2\pi^2}V+O(T^{1/2}+(\sum_{p\le T^{1/2}}p^{-1/2})^2)=\frac{T}{2\pi^2}V+O(T^{1/2})$. For $S_X^{(2)}(t)=-\frac{1}{2\pi}\sum_{p\le X}p^{-1}\sin(t\log p^2)$ the same lemma gives $\int(S_X^{(2)})^2\ll T\sum_p p^{-2}+\#\{p\le X\}+(\sum_{p\le X}p^{-1})^2\ll T$. Finally $|S_X^{(\ge3)}|\le\frac1\pi\sum_p\sum_{m\ge3}\frac{1}{mp^{m/2}}\le\frac1\pi\sum_p\frac{p^{-3/2}}{1-p^{-1/2}}<\infty$, uniformly in $X$. Expanding $\mathcal{W}^2$ and bounding the cross terms by Cauchy–Schwarz gives (5.6).

For (5.7) use $(y_1+y_2+y_3)^4\le 27(y_1^4+y_2^4+y_3^4)$. Let $\mathcal{D}_1(t)=\sum_{p\in\mathcal{J}}p^{-1/2}p^{-it}$, so that $|\mathcal{W}_1|\le|\mathcal{D}_1|/\pi$ and $\mathcal{D}_1^2=\sum_n\eta_n n^{-it}$ with $\eta_{p^2}=p^{-1}$, $\eta_{p_1p_2}=2(p_1p_2)^{-1/2}$ for $p_1<p_2$, all $n\le T$. Theorem 3.6 gives $\int|\mathcal{D}_1|^4=\sum|\eta_n|^2(T+O(n))\ll T\sum|\eta_n|^2\le T(1+2V^2)$. For $S_X^{(2)}$ let $\mathcal{D}_2(t)=\sum_{p\le X}p^{-1}(p^2)^{-it}$, so that $|S_X^{(2)}|\le|\mathcal{D}_2|/2\pi$ and $\mathcal{D}_2^2=\sum_n\eta_n' n^{-it}$ over $n=(p_1p_2)^2$ with $|\eta_n'|\le 2(p_1p_2)^{-1}$; then $\sum|\eta_n'|^2\ll 1$ and $\sum n|\eta_n'|^2\le 4\#\{p\le X\}^2\le 4X^2\le 4T$. $S_X^{(\ge3)}$ is bounded.

For (5.8), $S_X'(t)=\sum_{n\in A_X}a_n\log n\cos(t\log n)$ (termwise differentiation, Section 2.4). The terms $m=1$ have $|z_p|=\frac{\log p}{\pi\sqrt{p}}$; Lemma 5.8 and Theorem 3.7 give $\ll T\log X\sum_{p\le X}\frac{\log p}{p}+\sum_{p\le X}\log^2 p+(\sum_{p\le X}\frac{\log p}{\sqrt{p}})^2\ll T(1+\log X)^2+X\log X$, and $X\le T^{1/2}$. The terms $m=2$ have $|z_{p^2}|=\frac{\log p}{\pi p}$ and contribute $\ll T+X\log X+\log^2 X$. The terms $m\ge3$ are bounded by $\frac1\pi\sum_p\log p\sum_{m\ge3}p^{-m/2}<\infty$. Combine with $(y_1+y_2+y_3)^2\le 3(y_1^2+y_2^2+y_3^2)$. □

*5.2. Proof of Theorem 5.1*

*Proof. Integrability.* On $[T,2T]$, $S$ is bounded (Theorem 3.1) and piecewise continuous, and $S_X$, $S_X'$, $\varpi$ are continuous; all integrals below exist.

*Second and fourth moments of $E_X$.* Let $E_0=S-S_{Y_0}^{\mathrm{pr}}$ and $E_X=S-S_X=E_0+\mathcal{W}$. Theorem 3.5 and Hölder's inequality give

$$\int_T^{2T}E_0^4\,dt\ll T,\qquad\int_T^{2T}E_0^2\,dt\le T^{1/2}\Big(\int_T^{2T}E_0^4\,dt\Big)^{1/2}\ll T.\tag{5.9}$$

Let $\mathcal{E}_2=\int_T^{2T}E_X^2\,dt$. From (5.6), (5.9) and Cauchy–Schwarz,

$$\mathcal{E}_2=\frac{T}{2\pi^2}V+O\Big(T(1+\sqrt{V})\Big),\tag{5.10}$$

and from (5.7), (5.9) and $(y_1+y_2)^4\le 8(y_1^4+y_2^4)$,

$$\int_T^{2T}E_X^4\,dt\ll T(1+V)^2.\tag{5.11}$$

*Transfer to the ordinates.* Let $k_1, k_2$ be the smallest and largest indices in $\mathcal{I}_T$ ($\mathcal{I}_T \neq \emptyset$ for $T \geq T_0$) and apply (4.5). By Lemma 2.2 and (2.10), $|u_k(X)| \ll \log T + \sqrt{X} \ll T^{1/4}$, so the boundary terms and $u_{k_1}(X)^2 + u_{k_2}(X)^2$ are $O(T^{3/4})$. The intervals $[T, \gamma_{k_1}]$ and $[\gamma_{k_2}, 2T]$ have bounded length (Theorem 3.1), and on them $E_X^2(\mathcal{L} + |S_X'|) \ll T^{1/2} \cdot T^{1/4}$. Since $|\varpi| \ll 1/T$, replacing $(E_X + \varpi/\pi)^2$ by $E_X^2$ costs $\ll T^{-1} \int_T^{2T} (|E_X| + 1)(\mathcal{L} + |S_X'|)\, dt \ll (1+V)^{1/2} \log T$ by Cauchy–Schwarz, (5.10) and (5.8). Hence

$$\sum_{k \in \mathcal{I}_T} u_k(X)^2 = \int_T^{2T} E_X^2 \mathcal{L}\, dt + \int_T^{2T} E_X^2 S_X'\, dt - \frac{\#\mathcal{I}_T}{12} + O(T^{3/4}). \tag{5.12}$$

*Main term and derivative term.* Since $0 \leq \mathcal{L}(t) - \mathcal{L}(T) \leq \frac{\log 2}{2\pi}$ on $[T, 2T]$, the first integral is $\mathcal{L}(T)\mathcal{E}_2 + O(\mathcal{E}_2)$. By Cauchy–Schwarz, (5.11) and (5.8),

$$\left| \int_T^{2T} E_X^2 S_X'\, dt \right| \leq \left( \int_T^{2T} E_X^4 \right)^{1/2} \left( \int_T^{2T} S_X'^2 \right)^{1/2} \ll T(1+V)(1+\log X). \tag{5.13}$$

Dividing by $\#\mathcal{I}_T = T\mathcal{L}(T) + O(T)$ and using (5.10),

$$\frac{1}{\#\mathcal{I}_T} \sum_{k \in \mathcal{I}_T} u_k(X)^2 = \frac{V}{2\pi^2} + O\Big(1 + \sqrt{V} + \frac{1+V}{\log T} + (1+V)\, c\Big), \tag{5.14}$$

because $\log X = c \log T$. By Lemma 5.7, $(1+V)c \ll c(1 + \log \frac{1}{c}) \ll 1$ on $(0, \frac{1}{2}]$; $V \ll \log\log T$ since $c \geq \log 2 / \log T$, so $(1+V)/\log T = o(1)$. A second use of Lemma 5.7 replaces $\frac{V}{2\pi^2}$ by $\frac{1}{2\pi^2} \log \frac{1}{c} + O(1)$ and $\sqrt{V}$ by $\sqrt{\log(1/c)} + O(1)$, which proves (5.1). □

## 6. Theorem C: exact accounting for the one-step clock

For a finite nonempty set $I$ of indices and a prime power $n$, define the response statistics at the zeros and at the clock points,

$$\mathcal{R}_n^{\gamma}(I) = \frac{2}{a_n} \langle s_k \sin(\gamma_k \log n) \rangle_I, \qquad \mathcal{R}_n^{\tau}(I) = \frac{2}{a_n} \langle s_k \sin(\tau_k \log n) \rangle_I, \tag{6.1}$$

and, for a set $A$ of prime powers with $\sum_{n \in A} |a_n| < \infty$, the cross term

$$\mathrm{Cr}_A(I) = \langle S_A(\tau_k)^2 \rangle_I - v_A. \tag{6.2}$$

For a prime $p$, $2/a_p = -2\pi\sqrt{p}$. These are finite-window statistics of the zeros of the Riemann zeta function, defined without any hypothesis; $\mathrm{Cr}_A$ contains the finite-window diagonal discrepancies and all off-diagonal products and has no fixed sign.

**Theorem 6.1** (Finite-window accounting)**.** *Let $I$ be a finite nonempty set of indices and $A$ a set of prime powers with $\sum_{n \in A} |a_n| < \infty$.*

*(a) For every $k$ and $X \geq 2$,*

$$\begin{aligned} &r_k(X) = u_k(X) + \Big(S_X(\gamma_k) - S_X(\tau_k)\Big) + \chi_k, \\ &|\chi_k| \leq \frac{\delta_k^2}{2\pi \min(\gamma_k, \tau_k)}, \qquad |S_X(\gamma_k) - S_X(\tau_k)| \leq M_X |\delta_k|. \end{aligned} \tag{6.3}$$

*(b) For every real $\lambda$,*

$$\Big\langle s_k^2 - (s_k - \lambda S_A(\tau_k))^2 \Big\rangle_I = \sum_{n\in A} a_n^2 \Big(\lambda\, \mathcal{R}_n^\tau(I) - \frac{\lambda^2}{2}\Big) - \lambda^2\, \mathrm{Cr}_A(I). \tag{6.4}$$

*In particular, for the full local factors,*

$$\begin{aligned} &\langle s_k^2 - (s_k - S_X(\tau_k))^2\rangle_I = D_X(I) - \mathrm{Cr}_X(I), \\ &D_X(I) = \sum_{n\in A_X} a_n^2 \Big(\mathcal{R}_n^\tau(I) - \frac{1}{2}\Big) = 2\langle s_k S_X(\tau_k)\rangle_I - v_X, \end{aligned} \tag{6.5}$$

*with* $\mathrm{Cr}_X = \mathrm{Cr}_{A_X}$.

*Proof.* (a) By (2.11), $r_k(X) = \mathcal{L}(\tau_k)\delta_k - S_X(\tau_k)$. By (2.6), $\mathcal{L}(\tau_k)\delta_k = s_k + (\mathcal{L}(\tau_k) - \mathcal{L}(\xi_k))\delta_k$; the mean value theorem with $\mathcal{L}'(t) = 1/(2\pi t)$ and $|\tau_k - \xi_k| \le |\delta_k|$ gives $|\mathcal{L}(\tau_k) - \mathcal{L}(\xi_k)| \le |\delta_k|/(2\pi \min(\gamma_k, \tau_k))$. Insert $s_k = u_k(X) + S_X(\gamma_k)$. The last bound is the mean value theorem with (2.10).

(b) The series $S_A(\tau_k) = \sum_n a_n \sin(\tau_k \log n)$ converges absolutely, and so does the double series for $S_A(\tau_k)^2$; since $I$ is finite, the mean over $I$ may be exchanged with these sums. Expanding, the left side is $2\lambda\langle s_k S_A(\tau_k)\rangle_I - \lambda^2\langle S_A(\tau_k)^2\rangle_I$, with $\langle s_k S_A(\tau_k)\rangle_I = \sum_n a_n \langle s_k \sin(\tau_k \log n)\rangle_I = \frac{1}{2}\sum_n a_n^2 \mathcal{R}_n^\tau(I)$ and $\langle S_A(\tau_k)^2\rangle_I = v_A + \mathrm{Cr}_A(I)$. Taking $\lambda = 1$ and $A = A_X$ gives (6.5). □

*Remark* 6.2. (6.5) is an exact sign criterion for the gain of the full Euler phase at the clock points: the gain is positive exactly when the diagonal responses outweigh the cross term. No independence of the phases $\tau_k \log p$ is assumed. A threshold $\mathcal{R}_p^\tau = \frac{1}{2}$ describes a single diagonal prime contribution and does not determine the optimal cutoff when $\mathrm{Cr}_X$ and the prime powers matter.

*Remark* 6.3 (Optimal trigonometric correction). For a finite set $A$ of frequencies let $\varsigma_n(k) = \sin(\tau_k \log n)$, $\mathbf{Q} = (\langle \varsigma_{n_1}\varsigma_{n_2}\rangle_I)_{n_1,n_2\in A}$, $\mathbf{b} = (\langle s_k \varsigma_n(k)\rangle_I)_{n\in A}$ and $\Upsilon_{\boldsymbol{\lambda}} = \sum_n \lambda_n \varsigma_n$. Ordinary least squares gives $\langle (s_k - \Upsilon_{\boldsymbol{\lambda}}(k))^2\rangle_I = \langle s_k^2\rangle_I - 2\boldsymbol{\lambda}^{\mathsf{T}}\mathbf{b} + \boldsymbol{\lambda}^{\mathsf{T}}\mathbf{Q}\boldsymbol{\lambda}$, minimized by $\boldsymbol{\lambda} = \mathbf{Q}^\dagger\mathbf{b}$ with gain $\mathbf{b}^{\mathsf{T}}\mathbf{Q}^\dagger\mathbf{b}$; for prime powers $b_n = \frac{1}{2}a_n\mathcal{R}_n^\tau(I)$. The Euler phase is the choice $\lambda_n = a_n$, and (6.4) measures what it loses through $\mathcal{R}_n^\tau \ne 1$ and $\mathrm{Cr}_A \ne 0$.

*Remark* 6.4 (Clock point against zero). For a prime power $n$, $\sin(\gamma_k \ell_n) - \sin(\tau_k \ell_n) = \int_0^1 \frac{d}{dy}\sin((\tau_k - y\delta_k)\ell_n)\, dy$, whence, exchanging the finite mean and the integral,

$$\mathcal{R}_n^\gamma(I) - \mathcal{R}_n^\tau(I) = -\frac{2\ell_n}{a_n}\int_0^1 \Big\langle s_k \delta_k \cos\Big((\tau_k - y\delta_k)\ell_n\Big)\Big\rangle_I\, dy. \tag{6.6}$$

For fixed $n$ and $I = \mathcal{I}_T$ the right side is $O(\langle |s_k\delta_k|\rangle_{\mathcal{I}_T}) = O(\log\log T/\log T)$ by (2.6) and Remark 3.10; for $\ell_n \asymp \log T$ the phase change $\delta_k\ell_n$ is of order one, the transport integral need not be small, and the computed responses do differ (Section 9); no asymptotic statement about this difference is proved.

*Remark* 6.5 (Fixed prime sets). For fixed $X$, $|\delta_k^X - \delta_k| \le \sigma_X/\mathcal{L}(\tau_k) \to 0$ and $|u_k(X) - s_k| \le \sigma_X$. A fixed set of primes changes neither $\limsup_k |\delta_k|$ nor the unboundedness of the defect.

## 7. Computations

### *7.1. Data and methods*

The ordinates are those of Odlyzko's table of the first $K = 2\,001\,052$ zeros of the Riemann zeta function, accurate to within $4 \cdot 10^{-9}$ [18], converted to IEEE double precision by correctly rounded decimal-to-binary conversion ($\gamma_K = 1\,132\,490.658714\ldots$). Spot checks against `mpmath.zetazero` at eight indices differ by at most $3.5 \cdot 10^{-10}$. The clock values $\tau_k$ are the double-precision numbers nearest to the exact roots of (2.5). They are computed by Newton's method in extended precision and certified for every $k$ by evaluating $N_0 - (k - \frac{1}{2})$ at the rounded value and at both neighbouring floating-point numbers; the 20 679 indices at which the root lies within the evaluation error of a rounding midpoint are decided with `mpmath` at 40 significant digits, and an independent `mpmath` recomputation at 19 900 indices, including the first and last 100, agrees bit for bit. The Euler phase uses full local factors, evaluated either as $-\pi^{-1} \sum_{p \le X} \arg(1 - p^{-1/2-it})$ with the principal argument, or by (2.9) truncated where $p^{-m/2} \le 10^{-17}$. All means are over complete windows, without subsampling or fitted parameters. Windows of ordinals are $(0, 10^3]$, $(10^3, 10^4]$, $(10^4, 10^5]$, $(10^5, 5 \cdot 10^5]$, $(5 \cdot 10^5, 10^6]$, $(10^6, K]$; windows of height are $(T, 2T]$.

### *7.2. The clock and the ordinates*

Figure 1 shows $\delta_k = \tau_k - \gamma_k$ for $k \le 2000$ together with the residual of the one-step clock with $\mathcal{P} = \{2, 3, 5, 7\}$. The root mean square of $\delta_k$ decreases from 0.3587 on $(0, 10^3]$ to 0.1789 on $(10^6, K]$, of the order of magnitude of $(2 \log\log T)^{1/2} / \log T$ (Remark 3.10). The maxima per window decrease only from 0.9937 to 0.7174 (Table 4); all of them exceed 0.6331, which bounds only the limit superior. The finite-height envelope (3.3) holds at every index and is used to at most 15% of its size.

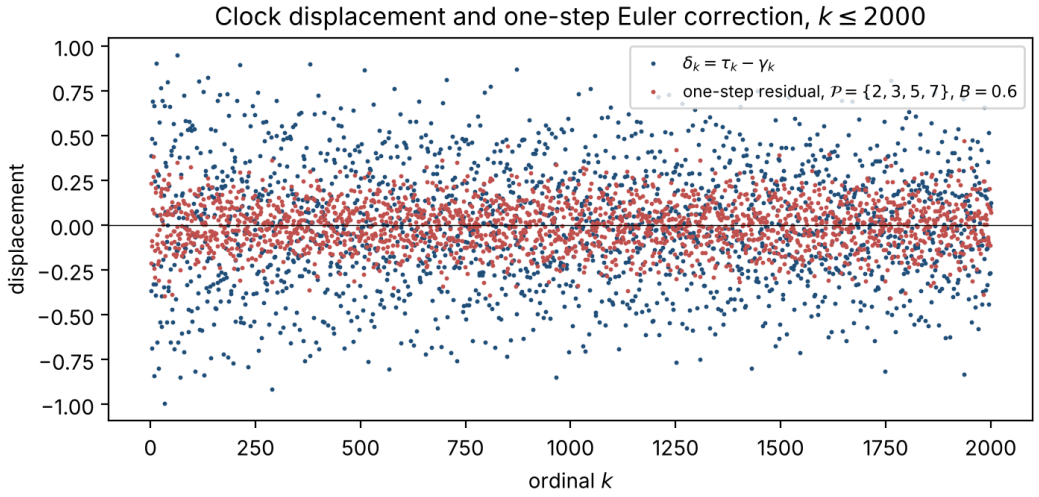



Figure 1: The displacement $\delta_k = \tau_k - \gamma_k$ of the RvM clock and the residual $\tau_k^{\mathcal{P}} - \gamma_k$ of the one-step clock with $\mathcal{P} = \{2, 3, 5, 7\}$ and saturation $B = 0.6$, for $k \le 2000$.

*7.3. Direct defects (Theorem B)*

For $T \in \{2^{14}, 2^{16}, 2^{18}, 2^{19}\}$ and $c \in \{\frac{1}{10}, \frac{1}{5}, \frac{1}{3}, \frac{1}{2}\}$ we compute $u_k(X)$ at the true ordinates with $X = T^c$ constant on $(T, 2T]$; the windows contain 21 519, 100 536, 459 982 and 977 804 ordinates. Smaller exponents give $X < 2$ at these heights and are excluded, as in Theorem 5.1.

Table 2: Mean squares of the defect $u_k(X)$ and of the one-step residual $r_k(X)$ over $T < \gamma_k \le 2T$, $X = T^c$.

| $T$ | $c$ | $X$ | $\#\{p \le X\}$ | $\langle u_k(X)^2 \rangle$ | $\langle r_k(X)^2 \rangle$ | $\frac{1}{2\pi^2} \log \frac{1}{c}$ |
|---|---|---|---|---|---|---|
| $2^{14}$ | 1/10 | 2.64 | 1 | 0.066738 | 0.066312 | 0.116650 |
| $2^{14}$ | 1/5 | 6.96 | 3 | 0.039487 | 0.038425 | 0.081535 |
| $2^{14}$ | 1/3 | 25.40 | 9 | 0.020174 | 0.018347 | 0.055656 |
| $2^{14}$ | 1/2 | 128.00 | 31 | 0.010929 | 0.010701 | 0.035115 |
| $2^{16}$ | 1/10 | 3.03 | 2 | 0.056337 | 0.055961 | 0.116650 |
| $2^{16}$ | 1/5 | 9.19 | 4 | 0.039588 | 0.039045 | 0.081535 |
| $2^{16}$ | 1/3 | 40.32 | 12 | 0.022282 | 0.021508 | 0.055656 |
| $2^{16}$ | 1/2 | 256.00 | 54 | 0.011533 | 0.012316 | 0.035115 |
| $2^{18}$ | 1/10 | 3.48 | 2 | 0.062696 | 0.062544 | 0.116650 |
| $2^{18}$ | 1/5 | 12.13 | 5 | 0.041377 | 0.041265 | 0.081535 |
| $2^{18}$ | 1/3 | 64.00 | 18 | 0.022737 | 0.022855 | 0.055656 |
| $2^{18}$ | 1/2 | 512.00 | 97 | 0.011983 | 0.013848 | 0.035115 |
| $2^{19}$ | 1/10 | 3.73 | 2 | 0.065599 | 0.065517 | 0.116650 |
| $2^{19}$ | 1/5 | 13.93 | 6 | 0.040530 | 0.040623 | 0.081535 |
| $2^{19}$ | 1/3 | 80.63 | 22 | 0.022911 | 0.023446 | 0.055656 |
| $2^{19}$ | 1/2 | 724.08 | 128 | 0.012187 | 0.014567 | 0.035115 |

At fixed $c$ the measured means are essentially independent of $T$, as Theorem 5.1 requires of a bounded quantity, and they decrease monotonically in $c$. Between $c = \frac{1}{2}$ and $c = \frac{1}{10}$ the increase of $\langle u_k^2 \rangle$ at $T = 2^{19}$ is 0.0534, against 0.0815 for the main term; at these heights the error term $O(1 + \sqrt{\log(1/c)})$ is not small compared with the main term, and the data do not determine a limit at fixed $c$. The one-step residual $r_k(X)$ agrees with $u_k(X)$ to within 3% for $c \le \frac{1}{5}$ and exceeds it at $c = \frac{1}{2}$ (0.0146 against 0.0122 at $T = 2^{19}$): this is the cost of evaluating $S_X$ at $\tau_k$ instead of $\gamma_k$ (Theorem 6.1(a)).

*7.4. Exact accounting (Theorem C)*

For $X \in \{7, 61, 401, 907, 1999\}$ and the five windows above $k = 1000$ we compute the gain $\langle s_k^2 - (s_k - S_X(\tau_k))^2 \rangle$, the diagonal term $D_X$ and the cross term $\mathrm{Cr}_X$ of (6.5). Table 3 lists three windows; Figure 3 shows all five.

The diagonal term $D_X$ increases with $X$ in every window and stays below $v_X$; the cross term $\mathrm{Cr}_X$ grows with $X$ at fixed height and decreases with the height at fixed $X$. The gain is therefore maximal at a cutoff that increases with the height: $X = 61$ on $(10^3, 10^4]$, between 401 and 1999 on $(10^5, 5 \cdot 10^5]$, and near 907 on $(10^6, K]$. No parameter is fitted, and the identity gain $= D_X - \mathrm{Cr}_X$ is exact.

*7.5. The one-step clock against the cutoff*

For comparison with [5, 14] we also report the root mean square and the maximum of $\tau_k^X - \gamma_k$ for 40 cutoffs $2 \le X \le 1999$ (Figure 4) and for five rules, with saturation

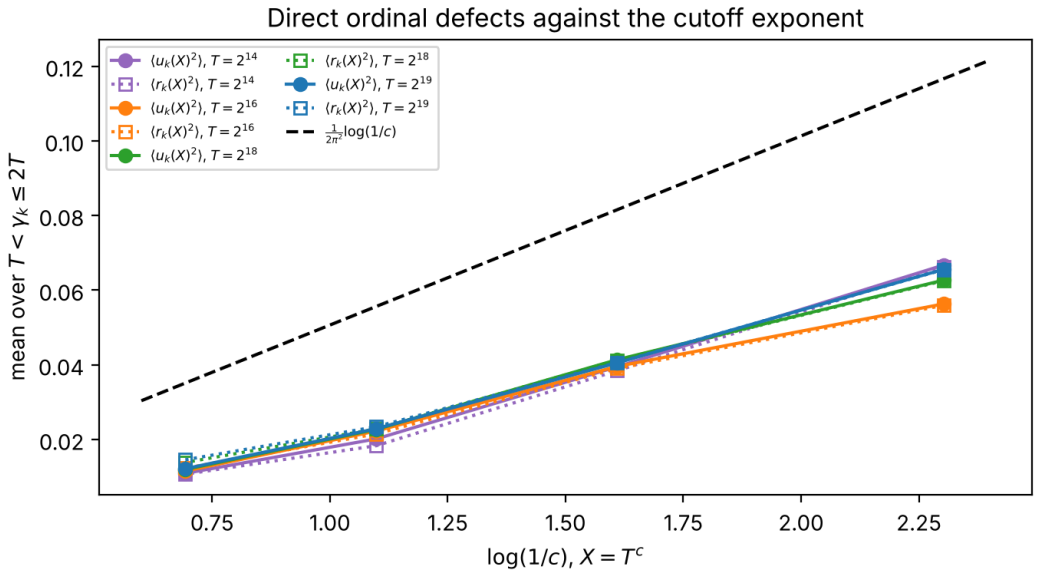



Figure 2: Mean square of $u_k(X)$ (filled markers) and of $r_k(X)$ (open markers) over $T < \gamma_k \le 2T$ against $\log(1/c)$, $X = T^c$, for the four heights of Table 2; dashed: $\frac{1}{2\pi^2}\log(1/c)$.

Table 3: Exact decomposition (6.5) of the one-step gain into the diagonal term $D_X$ and the cross term $\mathrm{Cr}_X$.

| window of $k$ | $\langle s_k^2\rangle$ | $X$ | gain | $D_X$ | $\mathrm{Cr}_X$ | $v_X$ |
|---|---|---|---|---|---|---|
| $(10^3, 10^4]$ | 0.085195 | 7 | 0.063224 | 0.063984 | 0.000760 | 0.066252 |
| | | 61 | 0.075868 | 0.085258 | 0.009390 | 0.093853 |
| | | 401 | 0.067945 | 0.097561 | 0.029616 | 0.111486 |
| | | 907 | 0.062337 | 0.100483 | 0.038146 | 0.117730 |
| | | 1999 | 0.055885 | 0.101829 | 0.045944 | 0.123202 |
| $(10^5, 5\cdot 10^5]$ | 0.106325 | 7 | 0.064185 | 0.064201 | 0.000016 | 0.066252 |
| | | 61 | 0.086367 | 0.086579 | 0.000212 | 0.093853 |
| | | 401 | 0.093276 | 0.096932 | 0.003657 | 0.111486 |
| | | 907 | 0.092779 | 0.099690 | 0.006911 | 0.117730 |
| | | 1999 | 0.091379 | 0.101921 | 0.010542 | 0.123202 |
| $(10^6, K]$ | 0.113224 | 7 | 0.064441 | 0.064444 | 0.000003 | 0.066252 |
| | | 61 | 0.087434 | 0.087478 | 0.000044 | 0.093853 |
| | | 401 | 0.098011 | 0.098495 | 0.000483 | 0.111486 |
| | | 907 | 0.098607 | 0.101389 | 0.002782 | 0.117730 |
| | | 1999 | 0.098325 | 0.103474 | 0.005149 | 0.123202 |

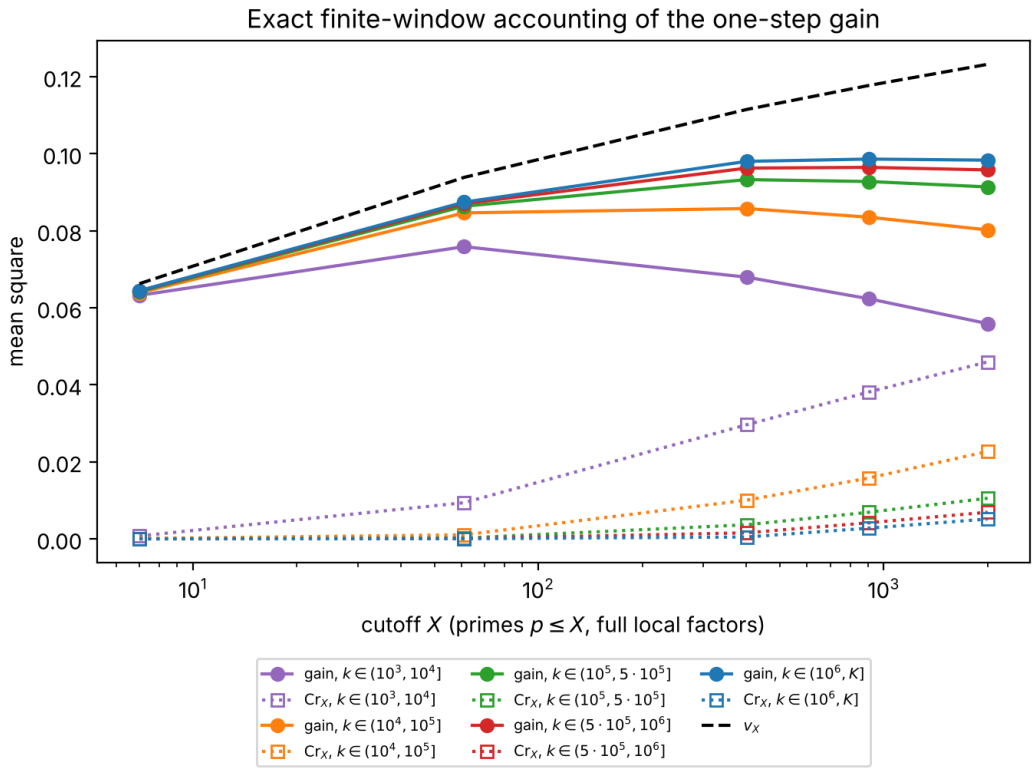



Figure 3: Gain of the one-step clock (filled markers) and cross term $\mathrm{Cr}_X$ (open markers) against the cutoff $X$, for five windows of ordinals; dashed: $v_X$.

$\tau_k - \operatorname{sgn}(y_k)\min(|y_k|, B)$, $y_k = S_X(\tau_k)/\mathcal{L}(\tau_k)$, where indicated (Table 4, Figure 5). The minimizing cutoff increases from 7 on $(0, 10^3]$ to 907 on $(10^6, K]$, in agreement with the accounting above, and a least-squares fit $\log X_{\mathrm{opt}} = \theta_{\mathrm{fit}} \log(\tau/2\pi)$ over the five upper windows gives $\theta_{\mathrm{fit}} = 0.5744$. This exponent is exploratory: the curves are flat near the minimum (in the top window the root mean square is 0.06447 at $X = 701$, 0.06427 at $X = 907$ and 0.06428 at $X = 1009$), and it is not asserted to be asymptotically optimal. A large fixed cutoff is harmful at small heights: $\mathcal{P} = \{p \le 1999\}$ gives the residual 3.247 at $k = 1$. The largest residual for $\mathcal{P} = \{2, 3, 5, 7\}$, at $k = 258\,318$, follows the close pair $\gamma_{258317} = 175\,681.6491$, $\gamma_{258318} = 175\,681.7159$ (normalized gap 0.109), at which $S(\gamma_{258318}) = 1.039$ and $\delta_{258318} = 0.638$.

Table 4: Root mean square, maximum of $|\tau_k^X - \gamma_k|$ and number of residuals $\ge \frac{1}{2}$ over all $K$ ordinates, and the same quantities in the top window $(10^6, K]$.

| rule | RMS | max | $\# \ge \frac{1}{2}$ | RMS, top | max, top | $\# \ge \frac{1}{2}$, top |
|---|---|---|---|---|---|---|
| bare clock $\tau_k$ | 0.188066 | 0.993711 | 11589 | 0.178900 | 0.717359 | 2971 |
| $\mathcal{P} = \{2, 3, 5, 7\}$, $B = 0.6$ | 0.120691 | 0.522347 | 1 | 0.117413 | 0.489293 | 0 |
| $\mathcal{P} = \{p \le 61\}$, $B = 0.6$ | 0.085635 | 0.457259 | 0 | 0.085354 | 0.400144 | 0 |
| $\mathcal{P} = \{p \le 1999\}$, no saturation | 0.070757 | 3.247397 | 156 | 0.064897 | 0.528267 | 7 |
| $X_k = \max(7, (\tau_k/2\pi)^{0.57})$, $B = 0.6$ | 0.066608 | 0.513886 | 2 | 0.064259 | 0.513886 | 1 |

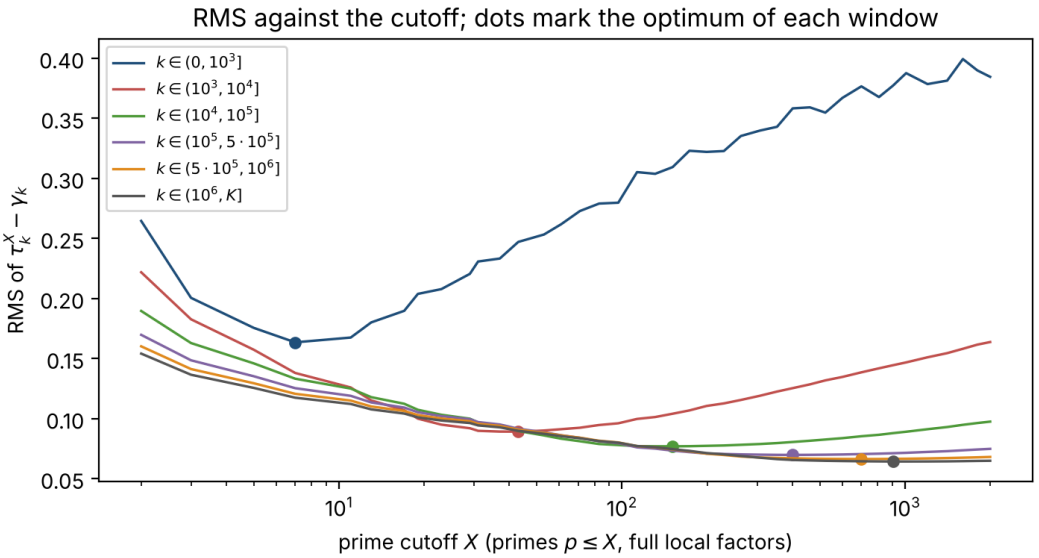



Figure 4: Root mean square of $\tau_k^X - \gamma_k$ (no saturation) against the cutoff $X$, by window of ordinals; dots mark the minimum in each window.

## 8. Conditional results

*Every statement of this section is either an unconditional identity or a rigorous implication from a named hypothesis. Nothing heuristic is used, and Theorems 4.1, 5.1 and 6.1 do not depend on this section.*

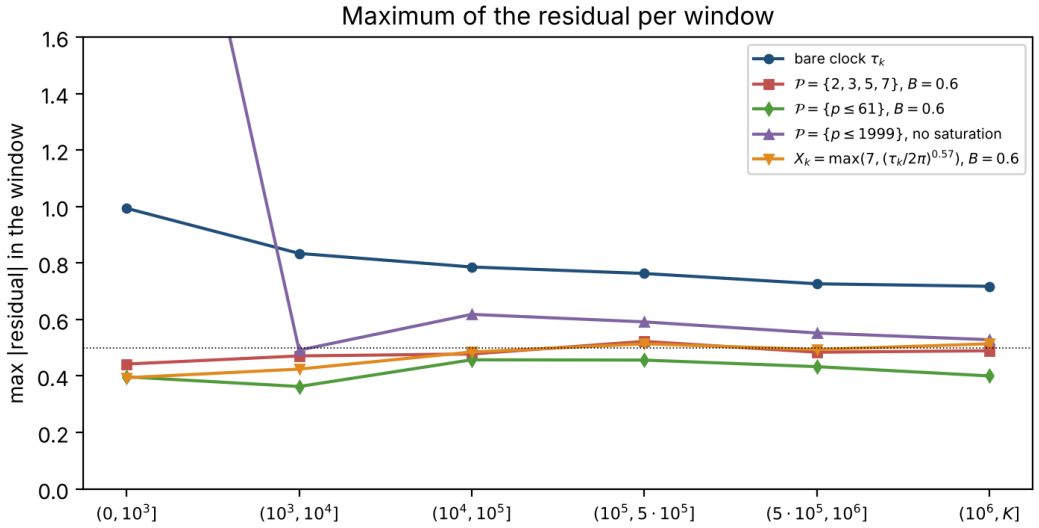



Figure 5: Maximum of $|\tau_k^X - \gamma_k|$ in each window of ordinals for the five rules of Table 4; dotted: $\frac{1}{2}$.

*8.1. Fixed frequencies under Hypothesis L*

**Hypothesis 8.1** (Hypothesis L: fixed-frequency Landau formula on the critical line)**.** For every fixed integer $n \ge 2$, $\sum_{0<\gamma\le t} n^{i\gamma} = -\frac{t}{2\pi}\frac{\Lambda(n)}{\sqrt{n}} + o(t)$, the zeros of the Riemann zeta function being counted with multiplicity.

**Proposition 8.2.** *Hypothesis L holds if* $\sum_{0<\gamma\le t} |\beta - \frac{1}{2}| = o(t)$.

*Proof.* $|n^\rho - \sqrt{n}\, n^{i\gamma}| = |n^\beta - n^{1/2}| \le n \log n\, |\beta - \frac{1}{2}|$ by the mean value theorem, since $0 < \beta < 1$; apply Theorem 3.9. □

**Lemma 8.3.** *For fixed real* $\omega \ne 0$, $\sum_{k\in\mathcal{I}_T} e^{i\omega\tau_k} \ll_\omega \log T$. *Hence for fixed integers* $n_1, n_2 \ge 2$, $\langle \sin(\tau_k \log n_1) \sin(\tau_k \log n_2)\rangle_{\mathcal{I}_T} \to \frac{1}{2}\mathbf{1}_{n_1=n_2}$.

*Proof.* Take $\omega > 0$ (conjugate otherwise) and $\Omega(x) = \omega\tau_x/2\pi$ for real $x \ge 1$. By (2.5), $\Omega'(x) = \omega/\log(\tau_x/2\pi)$ is continuous, positive and decreasing, and for $x \in [\min \mathcal{I}_T, \max \mathcal{I}_T]$ and $T \ge T_0(\omega)$ it lies in $[\omega/\log(T/\pi), \frac{1}{2}]$, so that $\|\Omega'\| \ge \omega/\log(T/\pi)$. $\mathcal{I}_T$ is a set of consecutive integers, and Theorem 3.8 gives the bound. For the second claim use $2\sin y_1 \sin y_2 = \cos(y_1 - y_2) - \cos(y_1 + y_2)$ with $\omega = \log(n_1/n_2)$ (if $n_1 \ne n_2$) and $\omega = \log(n_1 n_2)$, and $\#\mathcal{I}_T \asymp T \log T$. □

**Theorem 8.4** (Under Hypothesis L)**.** *Assume Hypothesis L. For every fixed integer* $n \ge 2$, $\langle s_k \sin(\gamma_k \log n)\rangle_{\mathcal{I}_T} \to \frac{a_n}{2}$, *and the same with* $\tau_k$ *in place of* $\gamma_k$; *thus* $\mathcal{R}_n^\gamma(\mathcal{I}_T), \mathcal{R}_n^\tau(\mathcal{I}_T) \to 1$ *for prime powers* $n$.

*Proof.* For $t > 14$ not an ordinate let $\mathcal{H}_n(t) = \int_{14}^t S(y) n^{iy} dy$; the integrand is bounded and piecewise continuous. As measures on $(14, t]$, $dS = dN - (\mathcal{L} + \varpi'/\pi)dy$, and $y \mapsto n^{iy}/(i\ell_n)$ is a $C^1$ primitive of $n^{iy}$; integration by parts for Lebesgue–Stieltjes integrals gives

$$\mathcal{H}_n(t) = \left[\frac{S(y)n^{iy}}{i\ell_n}\right]_{14}^t - \frac{1}{i\ell_n}\sum_{14<\gamma_k\le t} n^{i\gamma_k} + \frac{1}{i\ell_n}\int_{14}^t n^{iy}\Big(\mathcal{L} + \frac{\varpi'}{\pi}\Big)dy = \frac{i}{\ell_n}\sum_{0<\gamma_k\le t} n^{i\gamma_k} + O(\log t), \tag{8.1}$$

since $\gamma_1 > 14$, $S = O(\log t)$, $\int_{14}^{t} n^{iy}\mathcal{L}\,dy = O(\log t)$ by a second integration by parts ($\mathcal{L}'(y) = 1/(2\pi y)$), and $\varpi' = O(y^{-2})$. Hypothesis L gives $\mathcal{H}_n(t) = -i\frac{\Lambda(n)}{2\pi\sqrt{n}\,\ell_n}t + o(t)$. Integrating by parts once more, with $\frac{d}{dt}(t\mathcal{L}(t)) = \mathcal{L}(t) + \frac{1}{2\pi}$,

$$\int_T^{2T} S(t)n^{it}\mathcal{L}(t)\,dt = \Big[\mathcal{H}_n\mathcal{L}\Big]_T^{2T} - \int_T^{2T}\frac{\mathcal{H}_n(t)}{2\pi t}\,dt = -i\frac{\Lambda(n)}{2\pi\sqrt{n}\,\ell_n}\int_T^{2T}\mathcal{L}\,dt + o(T\log T). \quad (8.2)$$

Taking imaginary parts, $\int_T^{2T} S\sin(\ell_n t)\mathcal{L}\,dt = \frac{a_n}{2}\int_T^{2T}\mathcal{L}\,dt + o(T\log T)$. Apply Lemma 4.4 with $\psi(t) = \sin(\ell_n t)$ on $[t_1, t_2]$, where $t_1 \in [T, T + 1/T]$ and $t_2 \in [2T, 2T + 1/T]$ are non-ordinates; the change of endpoints alters both sides by $O(\log^2 T)$. The $\varpi$-term is $O(\log T)$, the boundary term is $O(\log^2 T)$, and $\frac{1}{2}\int G^2|\psi'| \ll \int_T^{2T} S^2 + O(1) \ll T\log\log T$ by Theorem 3.3. Divide by $\#\mathcal{I}_T = \int_T^{2T}\mathcal{L}\,dt + O(\log T)$. For $\tau_k$: $|\sin(\ell_n\tau_k) - \sin(\ell_n\gamma_k)| \le \ell_n|\delta_k|$, and by (2.6) and Remark 3.10, $\sum_{k\in\mathcal{I}_T}|s_k\delta_k| \ll (\log T)^{-1}\sum_{k\in\mathcal{I}_T} s_k^2 \ll \#\mathcal{I}_T\log\log T/\log T$. □

**Corollary 8.5** (Under Hypothesis L). *For a finite set $A$ of prime powers and $\lambda \in \mathbb{R}$, $\langle s_k^2 - (s_k - \lambda S_A(\tau_k))^2\rangle_{\mathcal{I}_T} \to (2\lambda - \lambda^2)v_A$, maximal at $\lambda = 1$.*

*Proof.* In (6.4), $\mathcal{R}_n^{\tau}(\mathcal{I}_T) \to 1$ by Theorem 8.4 and $\mathrm{Cr}_A(\mathcal{I}_T) \to 0$ by Lemma 8.3. □

The passage to the full local factors inside the limit needs uniformity in $n$, which is not proved. Figure 6 shows the gains of two fixed prime sets approaching $v_{\mathcal{P}}$ from below as the height increases.

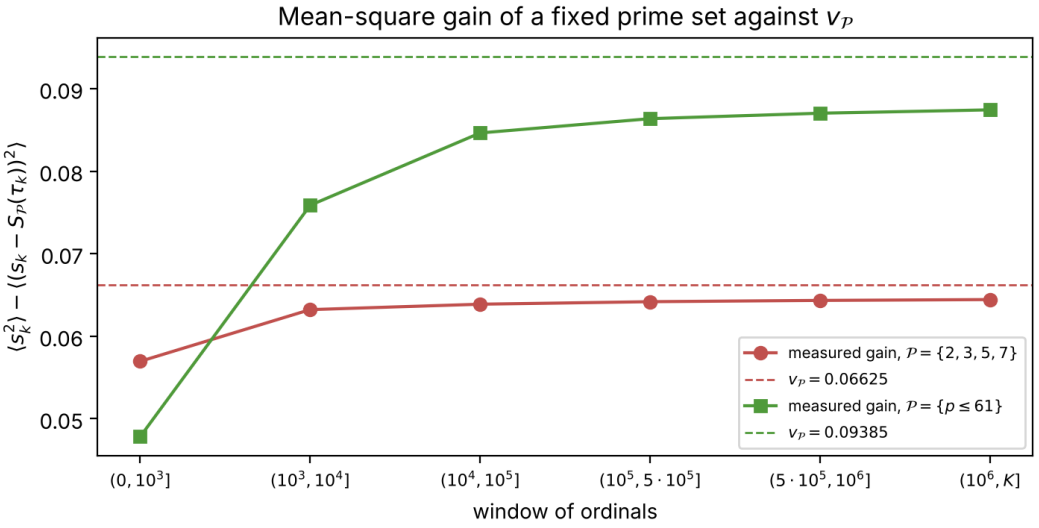



Figure 6: Mean-square gain $\langle s_k^2\rangle - \langle (s_k - S_{\mathcal{P}}(\tau_k))^2\rangle$ of two fixed prime sets $\mathcal{P}$ (full local factors) in each window of ordinals, and the constants $v_{\mathcal{P}}$ (dashed) of Corollary 8.5.

### *8.2. The response at the ordinates and its Fourier reduction*

Let $T, 2T$ be non-ordinates, $p$ a prime, and

$$\mathcal{A}_{T,p} = -\frac{2\pi\sqrt{p}}{\#\mathcal{I}_T}\int_T^{2T} G(t)\sin(t\log p)\mathcal{L}(t)\,dt, \qquad \mathcal{S}_{T,p} = \frac{1}{T}\int_T^{2T} S(t)^2 p^{it}\,dt. \quad (8.3)$$

**Proposition 8.6** (Unconditional separation of one- and two-point terms)**.**

$$\begin{aligned}\mathcal{R}_p^{\gamma}(\mathcal{I}_T) = \mathcal{A}_{T,p} + \frac{\pi\sqrt{p}\,T\log p}{\#\mathcal{I}_T}\Re\mathcal{S}_{T,p} - \frac{\pi\sqrt{p}}{\#\mathcal{I}_T}\big[G^2\sin(t\log p)\big]_T^{2T} \\ + \frac{\pi\sqrt{p}\log p}{\#\mathcal{I}_T}\int_T^{2T}(G^2-S^2)\cos(t\log p)\,dt.\end{aligned} \tag{8.4}$$

*If $\theta = \log p/L_T$ stays in a compact subset of $(0,1)$, the last two terms are $o(1)$ and*

$$\mathcal{R}_p^{\gamma}(\mathcal{I}_T) = \mathcal{A}_{T,p} + 2\pi^2\theta\sqrt{p}\,\big(1+O(1/L_T)\big)\Re\mathcal{S}_{T,p} + o(1). \tag{8.5}$$

*Proof.* Lemma 4.4 with $\psi(t) = \sin(t\log p)$ on $[T,2T]$, multiplied by $-2\pi\sqrt{p}/\#\mathcal{I}_T$, gives (8.4), because $\int_T^{2T} G^2\cos(t\log p)\,dt = T\Re\mathcal{S}_{T,p} + \int_T^{2T}(G^2-S^2)\cos(t\log p)\,dt$. On $[T,2T]$, $|G^2-S^2| \le 2|S||\varpi|/\pi + \varpi^2/\pi^2 \ll \log T/T$, so the last term is $O(\sqrt{p}\log p/T)$, and the boundary term is $O(\sqrt{p}\log^2 T/(T\log T))$. Both are $o(1)$ for $p \le T^{\theta_1}$ with $\theta_1 < 1$. Finally $\#\mathcal{I}_T = \frac{TL_T}{2\pi}(1+O(1/L_T))$. □

Let $f_T = S\mathbf{1}_{[T,2T]}$, let $\mathfrak{m}_T$ be its distributional derivative (a finite signed measure, since $f_T$ has compact support and bounded variation), and

$$Z_T(\omega) = \int_{\mathbb{R}} e^{i\omega t}\,d\mathfrak{m}_T(t), \qquad \widehat{f_T}(\omega) = \int_{\mathbb{R}} f_T(t)e^{i\omega t}dt. \tag{8.6}$$

**Theorem 8.7** (Unconditional shifted two-point representation)**.** *For non-ordinates $T, 2T$,*

$$Z_T(\omega) = \sum_{T<\gamma<2T}\nu(\gamma)e^{i\omega\gamma} - \int_T^{2T}\Big(\mathcal{L}(t) + \frac{\varpi'(t)}{\pi}\Big)e^{i\omega t}\,dt + S(T)e^{i\omega T} - S(2T)e^{2i\omega T}, \tag{8.7}$$

*$Z_T(\omega) = -i\omega\widehat{f_T}(\omega)$, so that $Z_T(0) = 0$ and $Z_T(\omega)/\omega$ extends continuously by $-i\widehat{f_T}$; and for every real $\omega_0$*

$$\int_T^{2T} S(t)^2 e^{i\omega_0 t}\,dt = -\frac{1}{2\pi}\int_{\mathbb{R}}\frac{Z_T(\omega)Z_T(\omega_0-\omega)}{\omega(\omega_0-\omega)}\,d\omega, \tag{8.8}$$

*the integral converging absolutely.*

*Proof.* On $(T,2T)$, $dS = dN - (\mathcal{L}+\varpi'/\pi)dt$ as measures, with atoms $\nu(\gamma)\delta_\gamma$; the cutoff adds $S(T)\delta_T$ and $-S(2T)\delta_{2T}$. This gives $\mathfrak{m}_T$ and (8.7). Since $f_T$ is of bounded variation with compact support, integration by parts gives $\int e^{i\omega t}d\mathfrak{m}_T = -i\omega\int f_T e^{i\omega t}dt$, the boundary terms vanishing. As $f_T \in L^1\cap L^2$ is real, Plancherel's identity applied to $f_T$ and $f_T e^{-i\omega_0 t}$ gives $\int f_T^2 e^{i\omega_0 t}dt = \frac{1}{2\pi}\int\widehat{f_T}(\omega)\widehat{f_T}(\omega_0-\omega)\,d\omega$, absolutely convergent since $\int|\widehat{f_T}(\omega)\widehat{f_T}(\omega_0-\omega)|d\omega \le \|\widehat{f_T}\|_2^2 = 2\pi\|f_T\|_2^2$. Substitute $\widehat{f_T} = iZ_T/\omega$. □

With $\omega_0 = \theta L_T = \log p$ and $\omega = \alpha L_T$ put

$$\mathcal{B}_T(\theta) = \int_{\mathbb{R}}\frac{Z_T(\alpha L_T)Z_T((\theta-\alpha)L_T)}{\alpha(\theta-\alpha)}\,d\alpha, \qquad \text{so that} \qquad \mathcal{S}_{T,p} = -\frac{\mathcal{B}_T(\theta)}{2\pi T L_T}. \tag{8.9}$$

Expanding the zero sums gives the weights $e^{i\alpha L_T\gamma}e^{i(\theta-\alpha)L_T\gamma'} = p^{i\gamma'}e^{i\alpha L_T(\gamma-\gamma')}$: $\mathcal{S}_{T,p}$ is a pair correlation of the zeros of the Riemann zeta function *shifted by a prime frequency*, which a stationary pair correlation (a function of $\gamma-\gamma'$ only) does not determine.

*8.3. The kernel $\mathfrak{K}_\theta$*

For $0 < \theta < 1$ define the continuous, piecewise linear function

$$\mathfrak{K}_\theta(\alpha) = \begin{cases} \theta, & \alpha \le -1, \\ \theta - 1 - \alpha, & -1 \le \alpha \le \theta - 1, \\ 0, & \theta - 1 \le \alpha \le 1, \\ \alpha - 1, & 1 \le \alpha \le 1 + \theta, \\ \theta, & \alpha \ge 1 + \theta, \end{cases} \tag{8.10}$$

equivalently, with $h(q) = \max(1 - |q|, 0)$,

$$\mathfrak{K}_\theta(\alpha) = 1 - h(\alpha) - h(\theta - \alpha) - h(\theta) + 2\max\Big(1 - \max(|\alpha|, |\theta - \alpha|, \theta), 0\Big), \tag{8.11}$$

as one checks on each of the five intervals.

**Proposition 8.8** (Integral and support of the kernel)**.** *For $0 < \theta < 1$,*

$$\int_{\mathbb{R}} \frac{\mathfrak{K}_\theta(\alpha)}{\alpha(\theta - \alpha)}\, d\alpha = -\frac{2\Phi(\theta)}{\theta}, \tag{8.12}$$

*the integral being absolutely convergent, and $\mathfrak{K}_\theta(\alpha) = 0$ whenever $|\alpha| + |\theta - \alpha| + \theta < 2$.*

*Proof.* $\mathfrak{K}_\theta$ is bounded and vanishes on $[\theta - 1, 1]$, which contains 0 and $\theta$; the integrand is bounded, continuous outside $[\theta - 1, 1]$ and $O(\alpha^{-2})$ as $|\alpha| \to \infty$, hence absolutely integrable. It is invariant under $\alpha \mapsto \theta - \alpha$, which maps $[1, \infty)$ onto $(-\infty, \theta - 1]$, so the integral is twice that over $[1, \infty)$. On $[1 + \theta, \infty)$, $\frac{\theta}{\alpha(\theta-\alpha)} = \frac{1}{\alpha} - \frac{1}{\alpha-\theta}$, whose integral is $\lim_{y\to\infty}\Big[\log\frac{\alpha}{\alpha-\theta}\Big]_{1+\theta}^{y} = -\log(1+\theta)$. On $[1, 1+\theta]$, $\frac{\alpha-1}{\alpha(\theta-\alpha)} = -\frac{1}{\theta\alpha} + \frac{\theta-1}{\theta(\theta-\alpha)}$, whose integral is $-\frac{\log(1+\theta)}{\theta} + \frac{\theta-1}{\theta}\log(1-\theta)$. The sum is $-\Phi(\theta)/\theta$; doubling gives (8.12). If $\alpha > 1$ then $|\alpha| + |\theta - \alpha| + \theta \ge \alpha + (\alpha - \theta) + \theta = 2\alpha > 2$, and if $\alpha < \theta - 1$ the same follows by the symmetry; elsewhere $\mathfrak{K}_\theta = 0$. □

*8.4. The moving-prime law under Hypotheses U and W*

**Hypothesis 8.9** (Hypothesis U: uniform one-particle input)**.** For every $0 < \theta_0 < \theta_1 < 1$, uniformly for primes with $\theta_0 \le \log p / L_T \le \theta_1$, $\mathcal{A}_{T,p} = 1 + o(1)$.

A sufficient condition is the uniform estimate $\sup_{T \le t \le 2T} |\sum_{T < \gamma_k \le t} p^{i\gamma_k} + \frac{(t-T)\log p}{2\pi\sqrt{p}}| = o(T \log p/\sqrt{p})$ in that range, by two integrations by parts as in the proof of Theorem 8.4. Theorem 3.9 concerns fixed bases and $p^\rho$, and does not give it.

**Hypothesis 8.10** (Hypothesis W: weighted arithmetic bridge)**.** Uniformly for primes with $\theta$ in any compact subset of $(0, 1)$,

$$\int_{\mathbb{R}} \frac{Z_T(\alpha L_T) Z_T((\theta - \alpha) L_T) + \mathcal{N}_T p^{-1/2} \mathfrak{K}_\theta(\alpha)}{\alpha(\theta - \alpha)}\, d\alpha = o\Big(\frac{T L_T}{\sqrt{p}}\Big), \tag{8.13}$$

with the continuous extensions of $Z_T(\omega)/\omega$ at $\omega = 0$.

**Proposition 8.11** (Moving-prime law under Hypotheses U and W)**.** *Under Hypothesis W,*

$$\mathcal{S}_{T,p} = -\frac{1}{2\pi^2\sqrt{p}}\frac{\Phi(\theta)}{\theta} + o(p^{-1/2}). \tag{8.14}$$

*Under Hypotheses U and W, or U and the real part of* (8.14)*, uniformly for $\theta$ in compact subsets of $(0,1)$,*

$$\mathcal{R}_p^{\gamma}(\mathcal{I}_T) = 1 - \Phi(\theta) + o(1). \tag{8.15}$$

*Conversely, under Hypothesis U,* (8.15) *is equivalent to the real part of* (8.14)*. Nothing is asserted about $\mathcal{R}_p^{\tau}$.*

*Proof.* The integral of the kernel term in (8.13) converges absolutely (Proposition 8.8); hence so does that of the first term, which is $\mathcal{B}_T(\theta)$, and by (8.12) $\mathcal{B}_T(\theta) = \frac{\mathcal{N}_T}{\sqrt{p}}\frac{2\Phi(\theta)}{\theta} + o(TL_T/\sqrt{p}) = \frac{TL_T}{\pi\sqrt{p}}\frac{\Phi(\theta)}{\theta} + o(TL_T/\sqrt{p})$; (8.9) gives (8.14). Then $\sqrt{p}\,\Re\mathcal{S}_{T,p} = O(1)$, so in (8.5) the factor $1 + O(1/L_T)$ contributes $o(1)$ and $2\pi^2\theta\sqrt{p}\,\Re\mathcal{S}_{T,p} = -\Phi(\theta) + o(1)$; with $\mathcal{A}_{T,p} = 1 + o(1)$ this is (8.15). Conversely, solve (8.5) for $\Re\mathcal{S}_{T,p}$. □

*Remark* 8.12. $\Phi(\theta) = 2\sum_{j\ge1}\frac{\theta^{2j}}{(2j-1)2j} = \theta^2 + \frac{\theta^4}{6} + \frac{\theta^6}{15} + \cdots$, so $1-\theta^2$ is only the quadratic truncation of (8.15), and $1-\Phi(\theta)$ vanishes at $\theta = 0.9036\ldots$. Hypotheses U and W are not consequences of Hypothesis L. The prime polynomial alone does not give (8.14): writing $S = S_Y^{\mathrm{pr}} + (S - S_Y^{\mathrm{pr}})$, the square $(S_Y^{\mathrm{pr}})^2$ has no exact Fourier frequency $\log p$, and Selberg's estimate gives only $|T^{-1}\int_T^{2T}(S - S_Y^{\mathrm{pr}})^2 p^{it}dt| \ll 1$, whereas (8.14) needs the error $o(p^{-1/2}) = T^{-\theta/2+o(1)}$.

## 9. Heuristic support and numerical evidence

*This section contains no statement about the Riemann zeta function that is used elsewhere. Subsection 9.1 proves a theorem about a random-matrix model; its relevance to the zeros of the Riemann zeta function is a heuristic.*

### *9.1. A determinantal model whose connected kernel is $\mathfrak{K}_\theta$*

On a circle of length $R_0$ let $\Pi_{\mathfrak{n}}$ be the orthogonal projection onto $\mathfrak{n}$ consecutive Fourier modes, and consider its determinantal point process, which has exactly $\mathfrak{n}$ points. For $q \in R_0^{-1}\mathbb{Z}$ let $\Xi_q$ be the centred linear statistic of $y \mapsto e^{2\pi i q y}$, so that $\Xi_0 = 0$ and $\overline{\Xi_q} = \Xi_{-q}$. Let $R_0, \mathfrak{n} \to \infty$ with $\mathfrak{n}/R_0 \to 1$, so that the local kernel tends to the sine kernel. For lattice $\theta > 0$ and real $\iota_1, \iota_2$ the tilt

$$d\mathbb{P}_{\iota_1,\iota_2} = \frac{\exp\{-\iota_1\Re\Xi_{-\theta} - \iota_2\Im\Xi_{-\theta}\}}{\mathbb{E}\exp\{-\iota_1\Re\Xi_{-\theta} - \iota_2\Im\Xi_{-\theta}\}}\,d\mathbb{P}_{0,0} \tag{9.1}$$

is a probability measure (the statistics are bounded), and differentiation at $(0,0)$, justified by boundedness, gives $(\partial_{\iota_1} + i\partial_{\iota_2})\operatorname{Cov}_{\iota_1,\iota_2}(\Xi_\alpha, \Xi_{\theta-\alpha})|_{(0,0)} = -\operatorname{cum}_3(\Xi_\alpha, \Xi_{\theta-\alpha}, \Xi_{-\theta})$, the third joint cumulant.

**Theorem 9.1** (Connected kernel of the model)**.** *Let lattice frequencies tend to $\alpha, \theta-\alpha, -\theta$ with $0 < \theta < 1$. Then $R_0^{-1}\operatorname{cum}_3(\Xi_\alpha, \Xi_{\theta-\alpha}, \Xi_{-\theta}) \to \mathfrak{K}_\theta(\alpha)$ and $R_0^{-1}\operatorname{cum}_2(\Xi_\theta, \Xi_{-\theta}) \to \theta$.*

*Proof.* For commuting multiplication operators $\mathsf{m}_1, \mathsf{m}_2, \mathsf{m}_3$, the third joint cumulant of the corresponding linear statistics of a determinantal process with projection kernel is [21]

$$\begin{aligned}\operatorname{cum}_3 = {} & \operatorname{Tr}(\mathsf{m}_1\mathsf{m}_2\mathsf{m}_3\Pi_{\mathfrak{n}}) - \operatorname{Tr}(\mathsf{m}_1\mathsf{m}_2\Pi_{\mathfrak{n}}\mathsf{m}_3\Pi_{\mathfrak{n}}) - \operatorname{Tr}(\mathsf{m}_1\mathsf{m}_3\Pi_{\mathfrak{n}}\mathsf{m}_2\Pi_{\mathfrak{n}}) - \operatorname{Tr}(\mathsf{m}_2\mathsf{m}_3\Pi_{\mathfrak{n}}\mathsf{m}_1\Pi_{\mathfrak{n}}) \\ & + \operatorname{Tr}(\mathsf{m}_1\Pi_{\mathfrak{n}}\mathsf{m}_2\Pi_{\mathfrak{n}}\mathsf{m}_3\Pi_{\mathfrak{n}}) + \operatorname{Tr}(\mathsf{m}_1\Pi_{\mathfrak{n}}\mathsf{m}_3\Pi_{\mathfrak{n}}\mathsf{m}_2\Pi_{\mathfrak{n}}),\end{aligned} \tag{9.2}$$

the third-order term of the expansion of $\log\det(\mathrm{Id} + \Pi_{\mathfrak{n}}(e^{\mathsf{m}} - 1)\Pi_{\mathfrak{n}})$. Multiplication by $e^{2\pi i q y}$ shifts a mode by $R_0 q$; a trace with two projections counts the overlap of two integer intervals of length $\mathfrak{n}$, and a trace with three their triple overlap. For the shifts $R_0\alpha$, $R_0(\theta - \alpha)$, $-R_0\theta$, whose sum is 0, the triple overlap has length $\max(\mathfrak{n} - R_0 \max(|\alpha|, |\theta - \alpha|, \theta), 0)$ for both cyclic orders, and the pair overlaps have lengths $R_0 h(\cdot) + o(R_0)$. Dividing by $R_0$ gives the right side of (8.11). The same count for two operators gives $R_0^{-1} \operatorname{cum}_2(\Xi_\theta, \Xi_{-\theta}) \to 1 - h(\theta) = \theta$. □

*Why Hypothesis W is plausible.* A window $[T, 2T]$ contains about $\mathcal{N}_T = TL_T/2\pi$ zeros of the Riemann zeta function at unit mean spacing after unfolding, and the one-particle prime amplitude in Hypothesis U is $-\mathcal{N}_T\theta/\sqrt{p}$. This equals the linear response of the tilt (9.1) with intensity $p^{-1/2}$, by the susceptibility $\theta$ of Theorem 9.1. If the zeros responded to the prime $p$ like the model, the connected product in (8.9) would be $-\mathcal{N}_T\mathfrak{K}_\theta(\alpha)/\sqrt{p}$, which is the content of Hypothesis W. This is an analogy, not an argument: stationary sine-kernel statistics do not determine the shifted correlation (8.9), and convergence on compact sets would not control the singular weight and the tails. Note that $\mathfrak{K}_\theta$ vanishes where $|\alpha| + |\theta - \alpha| + \theta < 2$ (Proposition 8.8), so the whole value $-2\Phi(\theta)/\theta$ comes from frequencies outside that region.

*9.2. Numerical evidence*

**Conjecture 9.2.** *For primes with $\theta = \log p/L_T$ in compact subsets of $(0, 1)$, $\mathcal{R}_p^\gamma(\mathcal{I}_T) = 1 - \Phi(\theta) + o(1)$.*

For all 2262 primes $p \le 20\,000$ we compute $\mathcal{R}_p^\gamma$ and $\mathcal{R}_p^\tau$ on the five windows of ordinals above $k = 1000$, with $\theta = \log p/\operatorname{median}_k \log(\tau_k/2\pi)$, and compare the primes with $0 < \theta < 1$ with $1 - \Phi(\theta)$. Table 5 separates individual errors from bin means over $\theta$-bins of width 0.05 containing at least five primes; Table 6 lists the bins of the two top windows, and Figure 7 all bins.

Table 5: Errors of $\mathcal{R}_p^\gamma - (1 - \Phi(\theta))$ over the primes $p \le 20\,000$ with $\theta < 1$.

| window of $k$ | primes | max individual error | RMS error | max \|bin-mean error\| |
|---|---|---|---|---|
| $(10^3, 10^4]$ | 159 | 0.34557 | 0.09425 | 0.02637 |
| $(10^4, 10^5]$ | 901 | 0.44085 | 0.08511 | 0.02371 |
| $(10^5, 5 \cdot 10^5]$ | 2262 | 0.27857 | 0.06429 | 0.02972 |
| $(5 \cdot 10^5, 10^6]$ | 2262 | 0.31682 | 0.06086 | 0.03531 |
| $(10^6, K]$ | 2262 | 0.16425 | 0.03309 | 0.01951 |

The bin means follow $1 - \Phi(\theta)$ and separate from $1 - \theta^2$ for $\theta \ge 0.6$. The residuals are not noise: in the two top windows every bin mean exceeds the candidate, by about 0.004 at $\theta \approx 0.3$ and 0.02–0.035 at $\theta \approx 0.8$–0.9, many standard errors, and the excess does not decrease between these two windows. The data are consistent with Conjecture 9.2 only

Table 6: Bin means of $\mathcal{R}_p^\gamma$ and $\mathcal{R}_p^\tau$ in the two top windows; "law" is the mean of $1-\Phi(\theta)$ over the same primes, s.e. the standard error of the mean difference.

| window of $k$ | $\theta$-bin | primes | $\mathcal{R}_p^\gamma$ | law | difference | s.e. | $\mathcal{R}_p^\tau$ |
|---|---|---|---|---|---|---|---|
| $(5\cdot 10^5, 10^6]$ | $[0.30, 0.35)$ | 6 | 0.8985 | 0.8933 | +0.0052 | 0.0003 | 0.8845 |
| | $[0.35, 0.40)$ | 8 | 0.8592 | 0.8549 | +0.0043 | 0.0001 | 0.8515 |
| | $[0.40, 0.45)$ | 13 | 0.8174 | 0.8127 | +0.0047 | 0.0002 | 0.8180 |
| | $[0.45, 0.50)$ | 22 | 0.7671 | 0.7619 | +0.0052 | 0.0004 | 0.7829 |
| | $[0.50, 0.55)$ | 33 | 0.7114 | 0.7059 | +0.0054 | 0.0004 | 0.7496 |
| | $[0.55, 0.60)$ | 54 | 0.6513 | 0.6443 | +0.0070 | 0.0007 | 0.7205 |
| | $[0.60, 0.65)$ | 86 | 0.5849 | 0.5753 | +0.0096 | 0.0008 | 0.6960 |
| | $[0.65, 0.70)$ | 142 | 0.5090 | 0.4979 | +0.0111 | 0.0009 | 0.6771 |
| | $[0.70, 0.75)$ | 234 | 0.4239 | 0.4103 | +0.0136 | 0.0012 | 0.6689 |
| | $[0.75, 0.80)$ | 380 | 0.3305 | 0.3123 | +0.0181 | 0.0017 | 0.6905 |
| | $[0.80, 0.85)$ | 630 | 0.2225 | 0.2018 | +0.0207 | 0.0021 | 0.7548 |
| | $[0.85, 0.90)$ | 645 | 0.1355 | 0.1002 | +0.0353 | 0.0034 | 0.8411 |
| $(10^6, K]$ | $[0.30, 0.35)$ | 7 | 0.8952 | 0.8909 | +0.0043 | 0.0001 | 0.8779 |
| | $[0.35, 0.40)$ | 12 | 0.8556 | 0.8518 | +0.0038 | 0.0002 | 0.8426 |
| | $[0.40, 0.45)$ | 16 | 0.8126 | 0.8085 | +0.0041 | 0.0003 | 0.8076 |
| | $[0.45, 0.50)$ | 27 | 0.7669 | 0.7629 | +0.0040 | 0.0002 | 0.7749 |
| | $[0.50, 0.55)$ | 48 | 0.7126 | 0.7076 | +0.0050 | 0.0003 | 0.7406 |
| | $[0.55, 0.60)$ | 77 | 0.6518 | 0.6453 | +0.0065 | 0.0004 | 0.7085 |
| | $[0.60, 0.65)$ | 129 | 0.5845 | 0.5761 | +0.0084 | 0.0006 | 0.6815 |
| | $[0.65, 0.70)$ | 222 | 0.5074 | 0.4974 | +0.0100 | 0.0007 | 0.6640 |
| | $[0.70, 0.75)$ | 368 | 0.4218 | 0.4099 | +0.0119 | 0.0009 | 0.6524 |
| | $[0.75, 0.80)$ | 627 | 0.3271 | 0.3120 | +0.0151 | 0.0011 | 0.6642 |
| | $[0.80, 0.85)$ | 718 | 0.2393 | 0.2198 | +0.0195 | 0.0016 | 0.7015 |

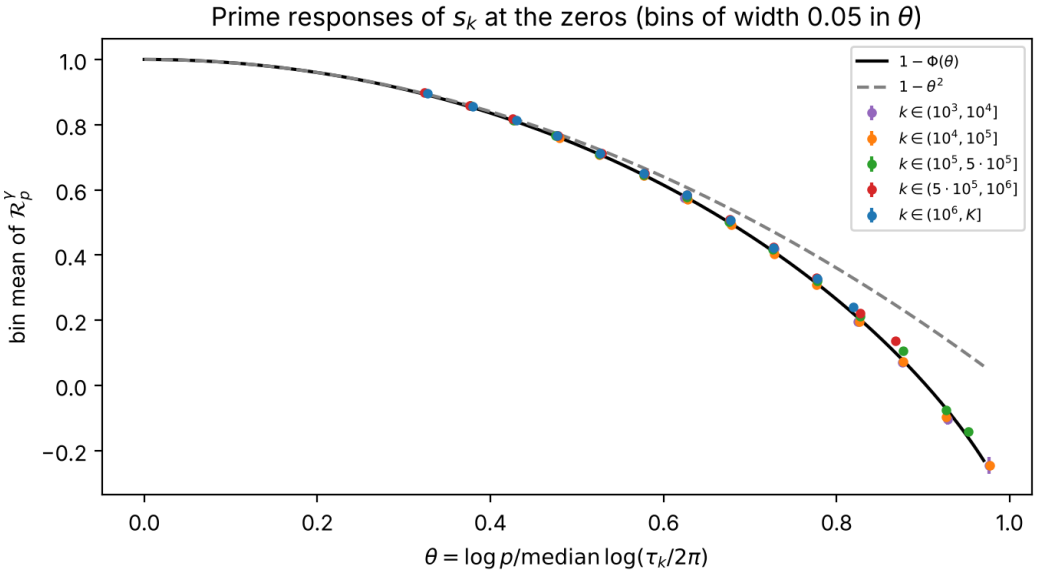



Figure 7: Bin means of $\mathcal{R}_p^\gamma$ (bars: standard error) against $1-\Phi(\theta)$ (solid) and $1-\theta^2$ (dashed), for five windows of ordinals.

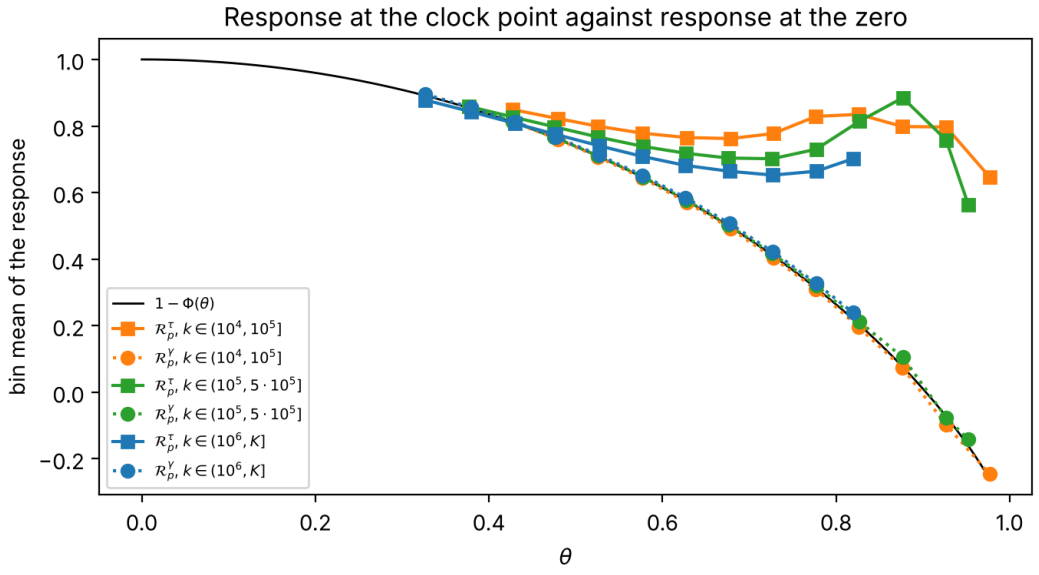



Figure 8: Bin means of the response at the clock points, $\mathcal{R}_p^\tau$ (squares), and at the zeros, $\mathcal{R}_p^\gamma$ (circles), for three windows of ordinals; solid: $1-\Phi(\theta)$.

together with a finite-height correction of a few hundredths whose decay is not seen below $\gamma = 1.13 \cdot 10^6$. The response at the clock points follows a different curve, with a plateau near 0.65–0.70 for $\theta \in [0.65, 0.80]$ in the top window (Figure 8), as Remark 6.4 allows.

## 10. Conclusions

### *10.1. What was known*

Before this work, the relation between the RvM clock and the ordinates of the zeros of the Riemann zeta function was known at two levels. Pointwise, $\delta_k = \tau_k - \gamma_k$ is bounded, with $\limsup |\delta_k| \leq 0.6331$ as an immediate consequence of the bound of Bellotti and Wong [1]. On average, $s_k$ is asymptotically Gaussian with variance $\frac{1}{2\pi^2} \log\log T$, by Selberg's formulae for the Gram displacement proved by Korolev [12, 11]: the clock is close to the zeros in $t$, but it misplaces almost every zero by an unbounded number of positions. Selberg's approximation [19, 7] replaces $S(t)$ by short prime sums in $t$-mean, and França–LeClair and LeClair [6, 14] write the equation of an individual zero with an Euler phase and a growing number of primes, with computations at a fixed number of primes.

### *10.2. What this paper adds*

1. *An exact dictionary between clock-type models and the zeros* (Theorem 4.1). For any $C^1$ model $F$ of the counting function, averages over the ordinates of the even powers of $k - \frac{1}{2} - F(\gamma_k)$ are exactly $dF$-averages of $(N - F)^{2j}$ in the continuous variable, corrected only by the sawtooth constant $\frac{1}{12}$ per zero (for $j = 1$) and two boundary terms. No hypothesis on the zeros, no short-interval counts and no monotonicity of $F$ are needed.

2. *The prime information needed by the clock* (Theorem 5.1). The corrected clock equation $N_0(t) + S_X(t) = k - \frac{1}{2}$ holds at the true ordinates with mean-square ordinal defect $\frac{1}{2\pi^2} \log \frac{\log T}{\log X}$ up to $O(1 + \sqrt{\log(\log T / \log X)})$. Each factor $e$ in $\log X$ removes $\frac{1}{2\pi^2}$ from the mean square. The corrected clock assigns the right position to almost all zeros up to a bounded error in mean square if and only if the primes reach a fixed power of the height; a fixed or slowly growing number of primes never does.

3. *An exact finite-height account of the computable clock* (Theorem 6.1 and Section 7). The gain of $\tau_k^X$ is the diagonal response minus a cross term, with no independence assumption, and the residual of $\tau_k^X$ differs from $u_k(X)$ by the explicit transport term $S_X(\gamma_k) - S_X(\tau_k)$. On the first 2 001 052 zeros the growth of the cross term, not a lack of prime information, sets the cutoff that is optimal at a given height.

4. *A sharp conditional target* (Sections 8 and 9). The prime content of $s_k$ at the zeros is reduced, exactly, to a pair correlation shifted by a prime frequency; under two named hypotheses it is $1 - \Phi(\theta)$. The data follow this curve up to a systematic excess of a few hundredths.

*10.3. Why a finer knowledge of $\tau_k - \gamma_k$ matters*

The clock is the deterministic part of the ordinates, and $\delta_k$ carries all the arithmetic. Statistics of $\delta_k$ and $s_k$ beyond the variance are statistics of the zeros of the Riemann zeta function that do not depend on an unfolding of the ordinates or on Gram intervals. Theorem 4.1 turns every question about weighted even moments of $k - \frac{1}{2} - F(\gamma_k)$ into a question about $N - F$ in $t$-mean. Any improvement in the $t$-mean description of $S(t)$ is therefore transferred to the zeros with the constant $\frac{1}{12}$ and nothing else: sharper forms of Selberg's approximation, prime polynomials longer than $T^{1/2}$, or mixed moments with prime powers would determine the constant left open at fixed $c$, and would extend Theorem 5.1 beyond $c = \frac{1}{2}$. In the other direction, the identities of Theorem 6.1 are exact at the heights where the zeros are known. The measured responses $\mathcal{R}_n^\tau$, $\mathcal{R}_n^\gamma$ and the cross terms $\mathrm{Cr}_X$ are therefore not approximations but finite-height invariants of the zeros, which any model of the zeros must reproduce. The prime responses of $s_k$ are an example: their bin means follow $1 - \Phi(\theta)$, a function that the stationary pair correlation of the zeros does not determine and that requires the shifted correlation (8.9). Their systematic deviation at the computed heights is a property of the zeros that no current asymptotic statement predicts.

*10.4. Future work*

**Question 10.1.** For fixed $c \in (0, \frac{1}{2}]$, do the bounded means of $u_k(T^c)^2$ converge, and to $(2\pi^2)^{-1}\log(1/c)$ plus which constant? This needs the mixed Selberg remainders and the prime-power terms, not only upper bounds.

**Question 10.2.** Can Theorem 5.1 be extended to $c > \frac{1}{2}$ (the proof uses $X^2 \le T$ in the fourth-moment step), and to a cutoff varying at every ordinate, such as $X_k = (\tau_k/2\pi)^c$?

**Question 10.3.** Can Hypotheses U and W be proved for moving primes, in the real or the complex form? What is the finite-height correction to $1 - \Phi(\theta)$ seen in Table 6? Is there a law for the response $\mathcal{R}^\tau$ at the clock points, with uniform control of the Gram matrix of Remark 6.3, that would predict the optimal cutoff of the computable clock?

Theorem 4.1 applies to any counting staircase and any $C^1$ model, in particular to the zeros of a primitive Dirichlet $L$-function; an analogue of Theorem 5.1 would additionally need Selberg's approximation for $L(s, \chi)$ [20, 7] and uniformity in the conductor.

## Declaration of competing interest

The author declares no competing interests.

## Data availability

All programs, the input-generation script with its certificates, the numerical outputs (CSV and JSON) and the figures are in the public GitHub repository `ordinal_prime_cutoff_v8` (link to be inserted). The ordinates are Odlyzko's table [18]; the two input files (arrays of 2 001 052 little-endian double-precision numbers) are regenerated from that table by `make_inputs_v8.py`. Their SHA-256 identifiers are `fafb9436ce6a...9e0015b3e945fe0ac` (ordinates) and `b4c6758f0aef...266091f713faa9504` (clock). The computations used Python 3.13.16, NumPy 2.5.3, SciPy 1.18.1, Matplotlib 3.11.2 and mpmath 1.3.0.

**Declaration of generative AI and AI-assisted technologies in the writing process**

During the preparation of this work the author used an AI assistant (Claude, Anthropic) to edit the text, check proofs and write analysis programs. The author reviewed and edited the content and takes full responsibility for it.

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